BS-M301

Probability and Statistics

Combined Module Notes

Modules 1 – 6

Semester III • ECE • 3 Credits

CGEC | B.Tech ECE (2023–27)

Disclaimer / Study Guide Notice:
These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

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Exam-Ready Notes

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Module 1

Probability Spaces, Conditional Probability and Independence

1. Probability Spaces, Conditional Probability and Independence

Modern probability theory is founded upon the rigorous measure-theoretic framework introduced by Andrey Kolmogorov in 1933. Rather than relying on heuristic ratios or subjective belief, probability is defined as a normalized measure assigned to events defined on a sample space.

Definition: Probability Space $(\Omega, \mathcal{F}, P)$

A **probability space** is a formal mathematical triplet $(\Omega, \mathcal{F}, P)$ comprising:

1. **Sample Space (Ω):** The set of all possible fundamental outcomes ω of a random experiment ($\omega \in \Omega$).
2. **Event Space (σ-algebra $\mathcal{F}$):** A collection of subsets of Ω satisfying three closure properties:
   - $\Omega \in \mathcal{F}$ (the certain event is included).
   - If $A \in \mathcal{F}$, then its complement $A^c = \Omega \setminus A \in \mathcal{F}$ (closure under complementation).
   - If $A_1, A_2, A_3, \cdots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (closure under countable unions).
3. **Probability Measure (P):** A real-valued mapping $P : \mathcal{F} \to [0, 1]$ satisfying Kolmogorov's Axioms.

Kolmogorov's Axioms of Probability

Every valid probability measure P on $(\Omega, \mathcal{F})$ must strictly obey:

$$\textbf{Axiom 1 (Non-negativity):} \quad P(A) \geq 0, \quad \forall A \in \mathcal{F} \tag{1}$$

$$\textbf{Axiom 2 (Total Certainty):} \quad P(\Omega) = 1 \tag{2}$$

$$\textbf{Axiom 3 (Countable Additivity):} \quad P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i), \quad \text{if } A_i \cap A_j = \emptyset \quad (\forall i \neq j) \tag{3}$$

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1.1 Fundamental Theorems Derived from Kolmogorov's Axioms

From these three primitive axioms, standard probability laws follow deductively:

1. **Empty Set Probability:** $P(\emptyset) = 0$, since $\Omega \cup \emptyset = \Omega$ and $\Omega \cap \emptyset = \emptyset \implies P(\Omega) + P(\emptyset) = P(\Omega) \implies P(\emptyset) = 0$.

2. **Complement Law:** For any event A, $A \cup A^c = \Omega$ and $A \cap A^c = \emptyset$, hence:

$$P(A^c) = 1 - P(A) \tag{4}$$

3. **Monotonicity:** If $A \subseteq B$, then $B = A \cup (B \setminus A)$ with $A \cap (B \setminus A) = \emptyset$. Thus $P(B) = P(A) + P(B \setminus A) \geq P(A)$, establishing:

$$A \subseteq B \implies P(A) \leq P(B) \tag{5}$$

4. **Addition Theorem for Arbitrary Events (Inclusion-Exclusion):** For any two events A and B:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{6}$$

For three events A, B, C:

$$\begin{aligned} P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) \\ + P(A \cap B \cap C) \end{aligned} \tag{7}$$

5. **Boole's Inequality (Union Bound):** For any countable collection of events $A_1, A_2, \ldots$:

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i) \tag{8}$$

Definition: Conditional Probability and Event Independence

Conditional Probability: The conditional probability of an event A given that event B has already occurred ($P(B) > 0$) is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \tag{9}$$

Independence of Events: Two events A and B are statistically **independent** if and only if:

$$P(A \cap B) = P(A) \cdot P(B) \iff P(A \mid B) = P(A) \quad (\text{provided } P(B) > 0) \tag{10}$$

Mutual Independence of n Events: Events $A_1, A_2, \ldots, A_n$ are mutually independent if for every sub-collection of k events ($2 \leq k \leq n$):

$$P\left(\bigcap_{j=1}^{k} A_{i_j}\right) = \prod_{j=1}^{k} P(A_{i_j}) \tag{11}$$

Crucial Exam Distinction: Pairwise independence ($P(A_i \cap A_j) = P(A_i)P(A_j)$ for all $i \neq j$) does **not** imply mutual independence!

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Law of Total Probability and Bayes' Theorem

Let $\{B_1, B_2, \dots, B_n\}$ form a **partition** of the sample space Ω such that:

$$\bigcup_{i=1}^{n} B_i = \Omega, \quad B_i \cap B_j = \emptyset \ (\forall i \neq j), \quad \text{and } P(B_i) > 0 \ \forall i \tag{12}$$

Law of Total Probability: For any arbitrary test event $A \subseteq \Omega$:

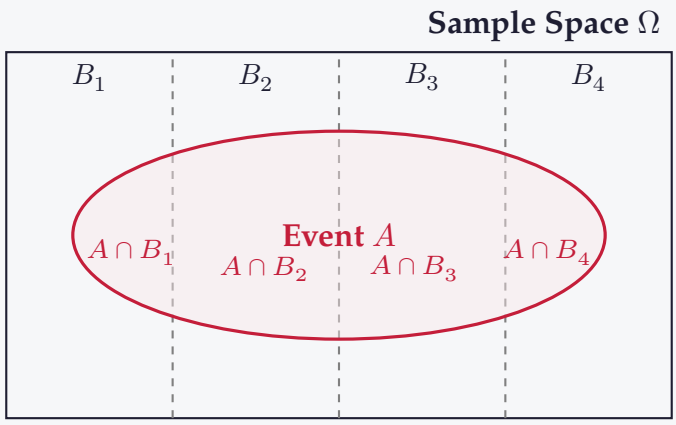
$$P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(B_i) \cdot P(A \mid B_i) \tag{13}$$

Bayes' Theorem (Posterior Probability): For any specific hypothesis B_k given evidence A:

$$P(B_k \mid A) = \frac{P(B_k \cap A)}{P(A)} = \frac{P(B_k) \cdot P(A \mid B_k)}{\sum_{i=1}^{n} P(B_i) \cdot P(A \mid B_i)} \tag{14}$$

where $P(B_k)$ is the *prior probability*, $P(A \mid B_k)$ is the *likelihood*, and $P(B_k \mid A)$ is the *posterior probability*.

TikZ Schematic: Sample Space Partitioning and the Law of Total Probability



$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) + P(A \cap B_4) = \textstyle\sum_{i=1}^{4} P(B_i)P(A \mid B_i)$$

2. Discrete Random Variables, PMF and CDF

Definition: Discrete Random Variable

A **random variable (RV)** X is a real-valued measurable function mapping the sample space into the real line:

$$X : \Omega \longrightarrow \mathbb{R} \tag{15}$$

A random variable is called **discrete** if its range $R_X = X(\Omega)$ is a finite or countably infinite set $\{x_1, x_2, x_3, \dots\}$.

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2.1 Probability Mass Function (PMF)

The **Probability Mass Function (PMF)** assigns probabilities directly to individual realization values x_i:

$$p_X(x_i) = P(X = x_i) = P(\{\omega \in \Omega : X(\omega) = x_i\}) \tag{16}$$

Every valid discrete PMF must satisfy two fundamental conditions:

1. **Non-negativity:** $p_X(x_i) \geq 0 \quad \forall x_i \in R_X$
2. **Unit Normalization:** $\sum_i p_X(x_i) = 1$

2.2 Cumulative Distribution Function (CDF)

The **Cumulative Distribution Function (CDF)**, denoted by $F_X(x)$, defines the probability that the random variable X takes on a value less than or equal to a specified threshold $x \in \mathbb{R}$:

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} p_X(x_i) \tag{17}$$

Mathematical Properties of the Discrete CDF

The CDF $F_X(x)$ of any discrete random variable satisfies:

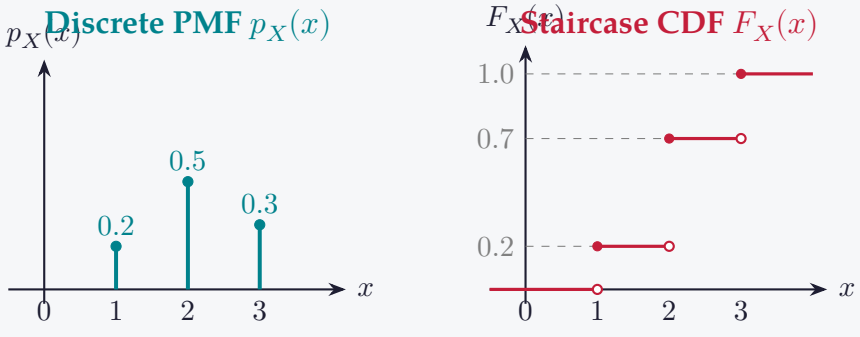
1. **Boundary Limits:** $\lim_{x \to -\infty} F_X(x) = 0 \quad \text{and} \quad \lim_{x \to +\infty} F_X(x) = 1$
2. **Monotonic Non-Decreasing:** If $a < b$, then $F_X(a) \leq F_X(b)$.
3. **Right-Continuity:** $\lim_{h \to 0^+} F_X(x + h) = F_X(x)$
4. **Step Discontinuities (Jump Heights):** At every point x_k with non-zero mass $p_X(x_k) > 0$, $F_X(x)$ exhibits a jump discontinuity of magnitude equal to the PMF:

$$P(X = x_k) = F_X(x_k) - F_X(x_k^-) = F_X(x_k) - \lim_{\epsilon \to 0^+} F_X(x_k - \epsilon) \tag{18}$$

5. **Interval Probabilities:**

$$P(a < X \leq b) = F_X(b) - F_X(a) \tag{19}$$
$$P(a \leq X \leq b) = F_X(b) - F_X(a^-) = F_X(b) - F_X(a) + P(X = a) \tag{20}$$

TikZ Illustration: Discrete PMF Stems and Corresponding Staircase CDF



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3. Expectation, Moments, Variance, Covariance and Correlation

Definition: Mathematical Expectation of a Discrete RV

Let X be a discrete random variable with PMF $p_X(x)$. The **expected value** (or mathematical expectation / first raw moment) of X is defined as:

$$E[X] = \mu = \sum_{x \in R_X} x \cdot p_X(x) \tag{21}$$

provided that the series converges absolutely: $\sum_x |x| p_X(x) < \infty$. If the series diverges, the expectation does not exist.

3.1 Properties of Mathematical Expectation

For arbitrary constants $a, b \in \mathbb{R}$ and measurable functions $g(X), h(X)$:

1. **Expectation of a Constant:** $E[c] = c \sum p_X(x) = c$.
2. **Linearity of Expectation:** For any random variables X and Y (independent or dependent!):
$$E[aX + bY + c] = aE[X] + bE[Y] + c \tag{22}$$
3. **Law of the Unconscious Statistician (LOTUS):** For any function $g(X)$:
$$E[g(X)] = \sum_{x \in R_X} g(x) \cdot p_X(x) \tag{23}$$
4. **Product Rule for Independent RVs:** If X and Y are independent:
$$E[X \cdot Y] = E[X] \cdot E[Y] \tag{24}$$

3.2 Moments: Raw Moments and Central Moments

- **r-th Raw Moment (μ'_r):** Expectation about the origin:
$$\mu'_r = E[X^r] = \sum_x x^r p_X(x), \quad \mu'_1 = \mu = E[X] \tag{25}$$
- **r-th Central Moment (μ_r):** Expectation about the mean μ:
$$\mu_r = E[(X - \mu)^r] = \sum_x (x - \mu)^r p_X(x) \tag{26}$$

  Specifically:

$$\mu_0 = E[(X - \mu)^0] = 1 \tag{27}$$
$$\mu_1 = E[X - \mu] = E[X] - \mu = 0 \quad \text{(Always zero!)} \tag{28}$$
$$\mu_2 = E[(X - \mu)^2] = \text{Var}(X) = \sigma^2 \tag{29}$$
$$\mu_3 = E[(X - \mu)^3] = \mu'_3 - 3\mu'_2\mu + 2\mu^3 \quad \text{(Skewness measure)} \tag{30}$$
$$\mu_4 = E[(X - \mu)^4] = \mu'_4 - 4\mu'_3\mu + 6\mu'_2\mu^2 - 3\mu^4 \quad \text{(Kurtosis measure)} \tag{31}$$

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Variance and Standard Deviation

The **variance** of X, denoted $\mathrm{Var}(X)$ or σ^2, measures the degree of dispersion around the mean:

$$\mathrm{Var}(X) = \sigma^2 = E\left[(X - E[X])^2\right] = E[X^2] - (E[X])^2 = \mu_2' - (\mu_1')^2 \tag{32}$$

Fundamental Properties of Variance:

1. $\mathrm{Var}(X) \geq 0$, and $\mathrm{Var}(X) = 0 \iff X = c$ (constant with probability 1).
2. $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$. Adding a constant shifts position without altering spread!
3. Standard deviation is $\sigma = \sqrt{\mathrm{Var}(X)}$, which retains the original units of X.

Definition: Covariance and Correlation Coefficient

Covariance: For two random variables X and Y with means μ_X and μ_Y, their **covariance** is:

$$\mathrm{Cov}(X,Y) = \sigma_{XY} = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - \mu_X\mu_Y \tag{33}$$

Pearson Correlation Coefficient (ρ): Normalized measure of linear dependence:

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y} = \frac{E[XY] - E[X]E[Y]}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} \tag{34}$$

3.3 Proof that $-1 \leq \rho \leq 1$

Rigorous Derivation: Cauchy-Schwarz Bound on Correlation

Consider the standardized variables $U = \frac{X-\mu_X}{\sigma_X}$ and $V = \frac{Y-\mu_Y}{\sigma_Y}$. Clearly $E[U] = E[V] = 0$, $E[U^2] = E[V^2] = 1$, and $E[UV] = \rho$. For any real scalar $t \in \mathbb{R}$, define the quadratic form:

$$Q(t) = E\left[(tU+V)^2\right] \geq 0 \quad \text{(since the square of a real quantity is non-negative)} \tag{35}$$

Expanding the quadratic expectation:

$$\begin{aligned} Q(t) &= E\left[t^2U^2 + 2tUV + V^2\right] \\ &= t^2E[U^2] + 2tE[UV] + E[V^2] \\ &= t^2(1) + 2t\rho + 1 \geq 0 \quad \forall t \in \mathbb{R} \end{aligned} \tag{36}$$

For a quadratic $At^2 + Bt + C \geq 0$ with $A = 1 > 0$ to remain non-negative for all real t, its discriminant $\Delta = B^2 - 4AC$ must be non-positive:

$$\Delta = (2\rho)^2 - 4(1)(1) \leq 0 \implies 4\rho^2 - 4 \leq 0 \implies \rho^2 \leq 1 \tag{37}$$

Taking square roots yields:

$$-1 \leq \rho \leq 1 \tag{38}$$

- $\rho = +1$: Perfect positive linear relationship ($Y = aX + b$ with $a > 0$).

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- $\rho = -1$: Perfect negative linear relationship ($Y = aX + b$ with $a < 0$).
- $\rho = 0$: X and Y are **uncorrelated**. Note: Independent $\implies$ Uncorrelated, but Uncorrelated does not imply Independent!

3.4 Variance of a Sum of Random Variables

For n random variables $X_1, X_2, \dots, X_n$:

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i) + 2 \sum_{1 \le i < j \le n} \operatorname{Cov}(X_i, X_j) \tag{39}$$

For two variables:

$$\operatorname{Var}(aX \pm bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) \pm 2ab \operatorname{Cov}(X, Y) \tag{40}$$

If X and Y are independent (or merely uncorrelated so that $\operatorname{Cov}(X, Y) = 0$):

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X - Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) \tag{41}$$

4. Bernoulli Trials, Multinomial, Poisson Approximation & Convolutions

4.1 Bernoulli Trials and Binomial Distribution

A **Bernoulli trial** is an experiment with exactly two mutually exclusive outcomes: *Success* (S) with probability p, and *Failure* (F) with probability $q = 1 - p$. In a sequence of n independent and identical Bernoulli trials:

Binomial Distribution Bin(n, p)

The random variable X, representing the total number of successes in n trials, follows the Binomial distribution:

$$P(X = k) = \binom{n}{k} p^k q^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}, \quad k \in \{0, 1, 2, \dots, n\} \tag{42}$$

Statistical parameters:

$$E[X] = np, \qquad \operatorname{Var}(X) = npq = np(1-p), \qquad \sigma_X = \sqrt{np(1-p)} \tag{43}$$

Since $q = 1 - p < 1$, the variance of a Binomial distribution is strictly less than its mean: $\operatorname{Var}(X) < E[X]$.

4.2 Multinomial Distribution

The **Multinomial distribution** generalizes the Binomial distribution to trials having $k \ge 2$ mutually exclusive outcomes $E_1, E_2, \dots, E_k$ with respective constant probabilities $p_1, p_2, \dots, p_k$ where $\sum_{j=1}^{k} p_j = 1$.

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Definition: Multinomial PMF

In n independent repetitions, let X_j denote the number of times event E_j occurs ($\sum_{j=1}^{k} X_j = n$). The joint PMF is:

$$P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k) = \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k} \tag{44}$$

where $x_j \in \{0, 1, \dots, n\}$ and $\sum_{j=1}^{k} x_j = n$. **Moments and Covariances:**

$$E[X_j] = np_j, \qquad \text{Var}(X_j) = np_j(1-p_j), \qquad \text{Cov}(X_i, X_j) = -np_i p_j \quad (i \neq j) \tag{45}$$

Key Insight: Covariance is strictly negative because an increase in occurrences of event E_i constrains the remaining trials available for event E_j.

4.3 Poisson Approximation to the Binomial Distribution (Law of Rare Events)

When the number of trials n is very large ($n \to \infty$) and the success probability p is very small ($p \to 0$) such that the average occurrence rate $\lambda = np$ remains a finite positive constant:

Rigorous Derivation: Poisson Limit Theorem

Starting from the Binomial PMF:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \tag{46}$$

Rearranging terms:

$$P(X = k) = \frac{\lambda^k}{k!} \cdot \left[\frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdots \frac{n-k+1}{n}\right] \cdot \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-k} \tag{47}$$

Taking the limit as $n \to \infty$ for fixed k:

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) = 1 \cdot 1 \cdots 1 = 1 \tag{48}$$

$$\lim_{n\to\infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda} \tag{49}$$

$$\lim_{n\to\infty} \left(1 - \frac{\lambda}{n}\right)^{-k} = (1-0)^{-k} = 1 \tag{50}$$

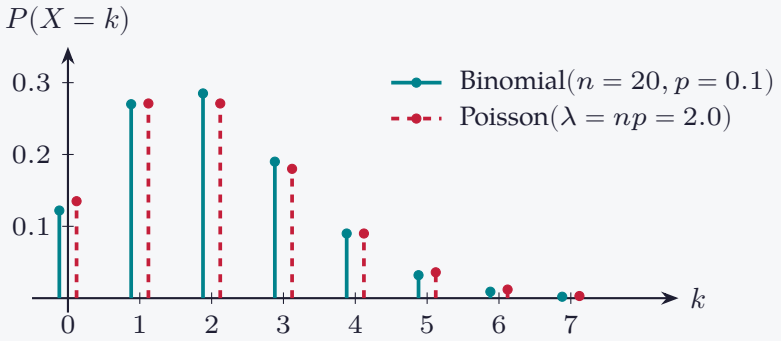
Multiplying these limits:

$$\lim_{n\to\infty} P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0, 1, 2, \dots \tag{51}$$

This is the PMF of the **Poisson distribution** with parameter $\lambda = np$. For Poisson, $E[X] = \text{Var}(X) = \lambda$.

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TikZ Plot: Binomial PMF ($n = 20, p = 0.1$) vs. Poisson PMF ($\lambda = np = 2.0$)



4.4 Infinite Sequences of Bernoulli Trials: Geometric Distribution

When independent Bernoulli trials are repeated indefinitely until the **first success** occurs:

$$P(X = k) = (1-p)^{k-1}p = q^{k-1}p, \qquad k \in \{1, 2, 3, \dots\} \tag{52}$$

Properties of the **Geometric distribution**:

$$E[X] = \frac{1}{p}, \qquad \text{Var}(X) = \frac{1-p}{p^2} = \frac{q}{p^2} \tag{53}$$

Memoryless Property: The geometric distribution is the *only* discrete distribution possessing memorylessness:

$$P(X > s + t \mid X > s) = P(X > t) \quad \forall s, t \in \mathbb{N} \tag{54}$$

4.5 Sums of Independent Random Variables (Discrete Convolution)

Let X and Y be independent discrete integer-valued random variables with PMFs $p_X(x)$ and $p_Y(y)$. The PMF of their sum $Z = X + Y$ is obtained via **discrete convolution**:

$$p_Z(z) = P(Z = z) = \sum_{k=-\infty}^{\infty} P(X = k, Y = z-k) = \sum_{k=-\infty}^{\infty} p_X(k) \cdot p_Y(z-k) = (p_X * p_Y)(z) \tag{55}$$

Reproductive (Convolution) Properties of Key Discrete Distributions

1. **Sum of Independent Binomials:** If $X \sim \text{Bin}(n_1, p)$ and $Y \sim \text{Bin}(n_2, p)$ share the same success parameter p:

$$Z = X + Y \sim \text{Bin}(n_1 + n_2, p) \tag{56}$$

2. **Sum of Independent Poissons:** If $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$:

$$Z = X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2) \tag{57}$$

Proof Sketch: Using the convolution formula:

$$P(Z = z) = \sum_{k=0}^{z} \frac{e^{-\lambda_1}\lambda_1^k}{k!} \cdot \frac{e^{-\lambda_2}\lambda_2^{z-k}}{(z-k)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{z!} \sum_{k=0}^{z} \binom{z}{k} \lambda_1^k \lambda_2^{z-k} = \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^z}{z!} \tag{58}$$

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5. Chebyshev's Inequality and Concentration Bounds

When the exact probability distribution of a random variable is unknown, Chebyshev's Inequality provides a guaranteed mathematical upper bound on the probability of significant deviations from the mean, requiring knowledge of only the mean μ and finite variance σ^2.

Markov's Inequality (Foundation for Chebyshev)

If Y is a non-negative random variable ($P(Y \geq 0) = 1$) with finite expectation $E[Y] < \infty$, then for any positive real constant $a > 0$:

$$P(Y \geq a) \leq \frac{E[Y]}{a} \tag{59}$$

Proof:

$$E[Y] = \sum_y y p_Y(y) = \sum_{y<a} y p_Y(y) + \sum_{y \geq a} y p_Y(y) \geq \sum_{y \geq a} y p_Y(y) \geq a \sum_{y \geq a} p_Y(y) = a P(Y \geq a) \tag{60}$$

Dividing both sides by $a > 0$ completes the proof.

Rigorous Derivation of Chebyshev's Inequality

Let X be any random variable with mean $\mu = E[X]$ and variance $\sigma^2 = \text{Var}(X) < \infty$. Define the non-negative random variable $Y = (X - \mu)^2 \geq 0$. Applying Markov's inequality to Y with threshold $a = \epsilon^2$ (where $\epsilon > 0$):

$$P\left(Y \geq \epsilon^2\right) \leq \frac{E[Y]}{\epsilon^2} \implies P\left((X-\mu)^2 \geq \epsilon^2\right) \leq \frac{E\left[(X-\mu)^2\right]}{\epsilon^2} \tag{61}$$

Since $(X - \mu)^2 \geq \epsilon^2 \iff |X - \mu| \geq \epsilon$, and $E[(X - \mu)^2] = \sigma^2$:

$$P(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2} \tag{62}$$

Substituting $\epsilon = k\sigma$ (where $k > 0$ represents the deviation in units of standard deviations):

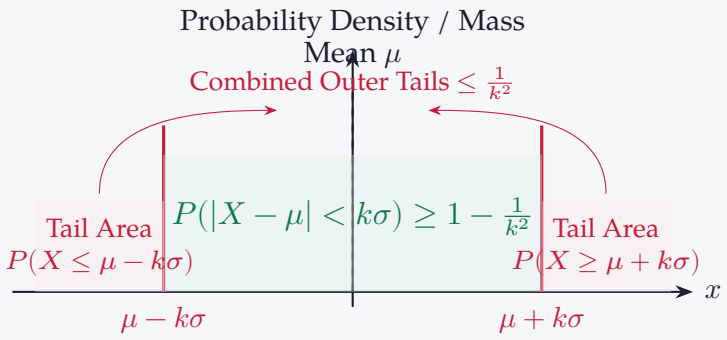
$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \tag{63}$$

Taking the complementary event yields the dual form:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \tag{64}$$

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TikZ Schematic: Geometric Interpretation of Chebyshev's Concentration Bound



5.1 Numerical Significance of Chebyshev's Bound

Spread (k)	**Interval** $[\mu - k\sigma, \mu + k\sigma]$	**Chebyshev Bound** $(P \geq 1 - 1/k^2)$	**Normal Distribution Comparison**
$k = 1$	$[\mu - \sigma, \mu + \sigma]$	$\geq 0\%$ **(Trivial bound)**	$\approx 68.27\%$
$k = 2$	$[\mu - 2\sigma, \mu + 2\sigma]$	$\geq 1 - 1/4 = 75.00\%$	$\approx 95.45\%$
$k = 3$	$[\mu - 3\sigma, \mu + 3\sigma]$	$\geq 1 - 1/9 = 88.89\%$	$\approx 99.73\%$
$k = 4$	$[\mu - 4\sigma, \mu + 4\sigma]$	$\geq 1 - 1/16 = 93.75\%$	$\approx 99.9937\%$

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Quick Revision — Module 1

Basic Probability, Discrete Random Variables and Inequalities | BS-M301

Key Formulas and Probability Identities at a Glance

Kolmogorov Axioms	$P(A) \geq 0$, $P(\Omega) = 1$, $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$ (disjoint)
Addition Law	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional Probability	$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, $P(A \cap B) = P(A) \cdot P(B)$ (Independence)
Total Probability	$P(A) = \sum_{i=1}^{n} P(B_i)P(A \mid B_i)$ (Partition $\{B_i\}$)
Bayes' Theorem	$P(B_k \mid A) = \frac{P(B_k)P(A\|B_k)}{\sum_{i=1}^{n} P(B_i)P(A\|B_i)}$
Discrete Expectation	$E[X] = \mu = \sum_x x p_X(x)$, $E[aX+bY] = aE[X]+bE[Y]$
Variance Formula	$\mathrm{Var}(X) = \sigma^2 = E[X^2] - (E[X])^2$, $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$
Covariance & Correlation	$\mathrm{Cov}(X,Y) = E[XY] - \mu_X\mu_Y$, $\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y} \in [-1,1]$
Binomial Distribution	$P(X=k) = \binom{n}{k}p^k q^{n-k}$, $\mu = np$, $\sigma^2 = npq$
Poisson Approximation	$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$, $\lambda = np$, $\mu = \sigma^2 = \lambda$
Chebyshev's Inequality	$P(\|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$, $P(\|X-\mu| < k\sigma) \geq 1 - \frac{1}{k^2}$

One-Line Definitions

- **Probability Space** $(\Omega, \mathcal{F}, P)$**:** Formal triplet of sample space, σ-algebra of events, and Kolmogorov measure.
- **σ-algebra ($\mathcal{F}$):** Event collection closed under complementation and countable unions containing Ω.
- **Statistical Independence:** Condition where $P(A \cap B) = P(A)P(B)$, giving zero mutual information.
- **Random Variable:** A measurable real-valued function $X : \Omega \to \mathbb{R}$ mapping outcomes to real numbers.
- **Probability Mass Function (PMF):** Discrete assignment $p_X(x) = P(X = x)$ with non-negativity and unit sum.
- **Cumulative Distribution Function (CDF):** Non-decreasing right-continuous probability $F_X(x) = P(X \leq x)$.
- **Law of Rare Events:** Poisson convergence of Binomial distribution when $n \to \infty, p \to 0$ with $\lambda = np$ constant.
- **Memoryless Property:** Invariance condition $P(X > s+t \mid X > s) = P(X > t)$, uniquely held by Geometric distribution.
- **Uncorrelated Variables:** Variables with zero covariance ($\mathrm{Cov}(X,Y) = 0, \rho = 0$), weaker than full independence.
- **Chebyshev's Bound:** Non-parametric inequality guaranteeing at least $(1 - 1/k^2)$ probability mass within $\mu \pm k\sigma$.

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Frequently Asked / Viva Questions (Part 1)

Q1. State Kolmogorov's Axioms of Probability and derive $P(\emptyset) = 0$.

Axiom 1: $P(A) \geq 0$. Axiom 2: $P(\Omega) = 1$. Axiom 3: Countable additivity for pairwise disjoint events. Since $\Omega = \Omega \cup \emptyset$ with $\Omega \cap \emptyset = \emptyset$, Axiom 3 yields $P(\Omega) = P(\Omega) + P(\emptyset) \implies P(\emptyset) = 0$.

Q2. Distinguish between pairwise independence and mutual independence with an example.

Pairwise independence requires $P(A_i \cap A_j) = P(A_i)P(A_j)$ for all pairs. Mutual independence additionally requires the product rule across all k-tuples up to all n events. In Bernstein's example (tossing two coins, A: heads on coin 1, B: heads on coin 2, C: both same), events are pairwise independent but not mutually independent.

Q3. State and prove the Law of Total Probability.

If $\{B_1, \ldots, B_n\}$ partitions Ω, then $A = A \cap \Omega = \bigcup_{i=1}^{n} (A \cap B_i)$. Because the partitions are disjoint, the slices $(A \cap B_i)$ are mutually exclusive. By additivity, $P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(B_i)P(A \mid B_i)$.

Q4. Under what physical conditions does the Binomial distribution tend to the Poisson distribution?

When the number of independent trials $n \to \infty$, the probability of success on each trial $p \to 0$, such that the expected average number of occurrences $\lambda = np$ remains a finite positive constant (the law of rare events).

Q5. Prove that for any random variable with finite variance, $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$.

$E[aX + b] = a\mu + b$. Then $\mathrm{Var}(aX + b) = E[((aX + b) - (a\mu + b))^2] = E[(a(X - \mu))^2] = E[a^2(X - \mu)^2] = a^2 E[(X - \mu)^2] = a^2 \mathrm{Var}(X)$. Adding a constant shifts the distribution without affecting spread.

Q6. Why is covariance negative for components of a Multinomial distribution?

In a multinomial distribution, $\sum_{j=1}^{k} X_j = n$ is fixed. If event i occurs more frequently ($X_i > E[X_i]$), fewer trials remain available for other events ($X_j < E[X_j]$), creating a deterministic negative association: $\mathrm{Cov}(X_i, X_j) = -np_ip_j < 0$.

Q7. Does zero correlation ($\rho = 0$) imply independence of two random variables?

No! Independence implies uncorrelatedness ($\mathrm{Cov}(X, Y) = 0 \implies \rho = 0$), but the converse is not true in general. For example, if $X \sim U[-1, 1]$ and $Y = X^2$, then $E[X] = 0$ and $E[XY] = E[X^3] = 0$, giving $\mathrm{Cov}(X, Y) = 0$ and $\rho = 0$, yet Y is completely deterministically dependent on X!

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Frequently Asked / Viva Questions (Part 2)

Q8. State and prove Chebyshev's Inequality using Markov's Inequality.

Markov's inequality for non-negative $Y \geq 0$ states $P(Y \geq a) \leq E[Y]/a$. Defining $Y = (X-\mu)^2 \geq 0$ and setting $a = k^2\sigma^2$: $P((X-\mu)^2 \geq k^2\sigma^2) \leq E[(X-\mu)^2]/(k^2\sigma^2) = \sigma^2/(k^2\sigma^2) = 1/k^2$. Since $(X-\mu)^2 \geq k^2\sigma^2 \iff |X-\mu| \geq k\sigma$, we obtain $P(|X-\mu| \geq k\sigma) \leq 1/k^2$.

Q9. What is the reproductive property of independent Poisson random variables?

The sum of two independent Poisson variables $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ is itself a Poisson random variable with parameter $\lambda_1 + \lambda_2$. This follows from the binomial expansion within the discrete convolution sum.

Q10. Explain the memoryless property of the Geometric distribution.

$P(X > s+t \mid X > s) = P(X > t)$ for integers $s, t \geq 1$. If s consecutive failures have already occurred, the probability of requiring at least t additional trials until the first success is identical to the initial probability of requiring at least t trials. The system has no memory of past failures.

Summary Matrix: Comprehensive Comparison of Discrete Probability Distributions

Distribution	Probability Mass Function (PMF)	Mean ($E[X]$)	Variance (σ^2)	Special Characteristic
Bernoulli	$P(X=x) = p^x(1-p)^{1-x},\ x \in \{0,1\}$	p	$p(1-p)$	Single trial; basic building block.
Binomial	$P(X=k) = \binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	$\sigma^2 < \mu$; independent repeated trials.
Poisson	$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!},\ k \in \mathbb{N}_0$	λ	λ	$\sigma^2 = \mu$; law of rare events.
Geometric	$P(X=k) = (1-p)^{k-1}p,\ k \in \mathbb{N}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	Only memoryless discrete distribution.
Multinomial	$P(X_1 = x_1, \dots) = \frac{n!}{\prod x_j!}\prod p_j^{x_j}$	np_j	$np_j(1-p_j)$	Negative covariance $\text{Cov} = -np_ip_j$.

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Module 2

Continuous Random Variables, PDF and CDF

1. Continuous Random Variables, PDF and CDF

A random variable is classified as **continuous** if its range consists of an entire interval or a union of intervals on the real line $\mathbb{R}$ (such as voltage, time, temperature, or noise amplitude). Because an interval contains uncountably infinitely many points, the probability that a continuous random variable assumes any exact single real value is strictly zero: $P(X = c) = 0$ for every $c \in \mathbb{R}$.

Definition: Probability Density Function (PDF)

A function $f_X : \mathbb{R} \to \mathbb{R}$ is called the **Probability Density Function (PDF)** of a continuous random variable X if for any interval $[a, b] \subseteq \mathbb{R}$:

$$P(a \leq X \leq b) = \int_a^b f_X(x)\, dx \tag{65}$$

Every valid PDF must strictly satisfy two axiomatic criteria:

$$\textbf{Non-negativity:} \quad f_X(x) \geq 0, \quad \forall x \in \mathbb{R} \tag{66}$$

$$\textbf{Total Normalization:} \quad \int_{-\infty}^{\infty} f_X(x)\, dx = 1 \tag{67}$$

Crucial Insight: The value $f_X(x)$ represents probability *density* per unit length, not probability itself! Consequently, $f_X(x)$ can exceed 1 over narrow intervals, provided the total area under the curve equals unity.

1.1 Cumulative Distribution Function (CDF)

The **Cumulative Distribution Function (CDF)** of a continuous random variable X is defined as the area under the PDF curve from $-\infty$ up to threshold x:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\, dt \tag{68}$$

By the Fundamental Theorem of Calculus, at every point where $f_X(x)$ is continuous, the PDF is the exact first derivative of the CDF:

$$f_X(x) = \frac{d}{dx} F_X(x) \tag{69}$$

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Key Properties of Continuous CDF and Interval Probabilities

1. **Continuous and Smooth:** Unlike discrete CDFs (which have jump discontinuities), the CDF $F_X(x)$ of a continuous random variable is everywhere continuous.
2. **Boundary Limits:** $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to+\infty} F_X(x) = 1$.
3. **Boundary Invariance of Interval Probabilities:** Since single-point probabilities are zero ($P(X = a) = P(X = b) = 0$):

$$P(a < X < b) = P(a \le X < b) = P(a < X \le b) = P(a \le X \le b)$$
$$= F_X(b) - F_X(a) = \int_a^b f_X(x)\,dx \tag{70}$$

4. **Tail (Survival) Function:** $R_X(x) = P(X > x) = 1 - F_X(x) = \int_x^\infty f_X(t)\,dt$.

1.2 Expectation, Variance and Moment Generating Function (MGF)

- **Mathematical Expectation (Mean):**

$$\mu = E[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx \quad \text{(provided } \int_{-\infty}^{\infty} |x| f_X(x)dx < \infty) \tag{71}$$

- **Variance (σ^2):**

$$\text{Var}(X) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x)\,dx = E[X^2] - (E[X])^2 \tag{72}$$

- **Moment Generating Function (MGF):** The MGF $M_X(t)$ is defined as the Laplace-type transform:

$$M_X(t) = E\left[e^{tX}\right] = \int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx \tag{73}$$

provided the integral converges in an open neighborhood around $t = 0$ ($|t| < h$).

- **Generation of Moments from MGF:** Expanding $e^{tX} = \sum_{r=0}^{\infty} \frac{t^r X^r}{r!}$:

$$M_X(t) = \sum_{r=0}^{\infty} \frac{t^r}{r!} E[X^r] = 1 + \mu_1' t + \frac{\mu_2' t^2}{2!} + \frac{\mu_3' t^3}{3!} + \cdots \tag{74}$$

Differentiating r times and setting $t = 0$:

$$\mu_r' = E[X^r] = \left.\frac{d^r M_X(t)}{dt^r}\right|_{t=0} \tag{75}$$

2. The Normal (Gaussian) Distribution

The **Normal distribution** (discovered by Carl Friedrich Gauss) is the central pillar of engineering probability, signal analysis, and statistics. By the Central Limit Theorem, the sum of a large number of independent random disturbances converges to a normal distribution, regardless of the underlying distribution of the individual components.

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Definition: Normal Distribution $\mathcal{N}(\mu, \sigma^2)$

A continuous random variable X has a **Normal distribution** with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$ (written $X \sim \mathcal{N}(\mu, \sigma^2)$) if its PDF is:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty \tag{76}$$

2.1 Mathematical Verification of the Normalizing Constant

To verify $\int_{-\infty}^{\infty} f_X(x)dx = 1$, let $I = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} dx$. Substitute the standard normal transform $z = \frac{x-\mu}{\sigma} \implies dx = \sigma dz$:

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2}\, dz \tag{77}$$

Let $J = \int_{-\infty}^{\infty} e^{-z^2/2} dz$. Converting J^2 to polar coordinates ($x = r\cos\theta, y = r\sin\theta, dxdy = rdrd\theta$):

$$J^2 = \int_{-\infty}^{\infty} e^{-x^2/2} dx \int_{-\infty}^{\infty} e^{-y^2/2} dy = \int_0^{2\pi} d\theta \int_0^{\infty} re^{-r^2/2}\, dr = 2\pi \left[-e^{-r^2/2}\right]_0^{\infty} = 2\pi(1) = 2\pi \tag{78}$$

Hence $J = \sqrt{2\pi}$, confirming $I = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1$.

Derivation: MGF and Moments of the Normal Distribution

For $X \sim \mathcal{N}(\mu, \sigma^2)$, the MGF is:

$$M_X(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}\, dx \tag{79}$$

Substituting $z = \frac{x-\mu}{\sigma} \implies x = \mu + \sigma z$ and $dx = \sigma dz$:

$$M_X(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\mu+\sigma z)} e^{-z^2/2}\, dz = e^{\mu t} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}(z^2 - 2\sigma t z)\right) dz \tag{80}$$

Completing the square in the exponent: $z^2 - 2\sigma tz = (z-\sigma t)^2 - \sigma^2 t^2$:

$$M_X(t) = e^{\mu t} e^{\sigma^2 t^2/2} \cdot \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-\sigma t)^2/2}\, dz}_{=1 \text{ (integral of standard normal density)}} = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right) \tag{81}$$

Moments via Derivatives of MGF:

$$M_X'(t) = (\mu + \sigma^2 t) e^{\mu t + \frac{1}{2}\sigma^2 t^2} \implies E[X] = M_X'(0) = \mu \tag{82}$$

$$M_X''(t) = \left[\sigma^2 + (\mu + \sigma^2 t)^2\right] e^{\mu t + \frac{1}{2}\sigma^2 t^2} \implies E[X^2] = M_X''(0) = \sigma^2 + \mu^2 \tag{83}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = (\sigma^2 + \mu^2) - \mu^2 = \sigma^2 \tag{84}$$

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2.2 Symmetry, Odd Moments and Inflection Points

1. **Symmetry about μ:** The bell curve is symmetric about the vertical line $x = \mu$: $f_X(\mu - x) = f_X(\mu + x)$. Consequently:

$$\text{Mean} = \text{Median} = \text{Mode} = \mu \tag{85}$$

2. **Vanishing of All Odd Central Moments:** By anti-symmetry of odd powers about zero:

$$\mu_{2k+1} = E[(X-\mu)^{2k+1}] = \int_{-\infty}^{\infty} (x-\mu)^{2k+1} f_X(x)\, dx = 0 \quad \forall k \in \mathbb{N}_0 \tag{86}$$

Specifically $\mu_1 = 0$ and $\mu_3 = 0$. Hence the skewness coefficient is identically zero: $\beta_1 = \gamma_1 = 0$.

3. **Even Central Moments and Kurtosis:**

$$\mu_{2k} = 1 \cdot 3 \cdot 5 \cdots (2k-1)\, \sigma^{2k} = \frac{(2k)!}{2^k k!} \sigma^{2k} \implies \mu_4 = 3\sigma^4 \tag{87}$$

The coefficient of kurtosis is: $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{3\sigma^4}{(\sigma^2)^2} = 3$ (Mesokurtic benchmark).

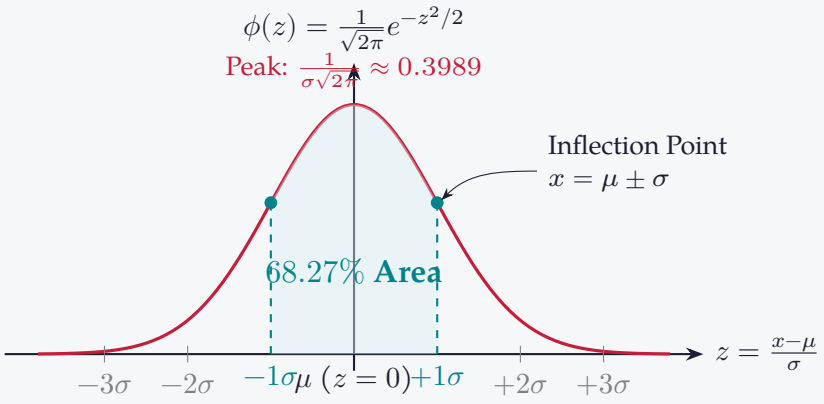
4. **Points of Inflection:** Finding where the concavity changes ($\frac{d^2 f_X}{dx^2} = 0$):

$$f_X'(x) = -\frac{x-\mu}{\sigma^2} f_X(x) \tag{88}$$

$$f_X''(x) = \left[\left(\frac{x-\mu}{\sigma^2} \right)^2 - \frac{1}{\sigma^2} \right] f_X(x) = 0 \implies (x-\mu)^2 = \sigma^2 \implies x = \mu \pm \sigma \tag{89}$$

The curve shifts from concave-downward to concave-upward at exactly $x = \mu \pm \sigma$.

TikZ Schematic: Standard Normal Gaussian Curve with Inflection Points and Area Rules



2.3 Standard Normal Distribution and $\Phi(z)$ Function

When $\mu = 0$ and $\sigma = 1$, the variable is denoted $Z \sim \mathcal{N}(0,1)$ with standard normal PDF $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and CDF:

$$\Phi(z) = P(Z \le z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-u^2/2}\, du \tag{90}$$

By symmetry:

$$\Phi(-z) = 1 - \Phi(z) \qquad \text{and} \qquad P(a \le X \le b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right) \tag{91}$$

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3. The Exponential Distribution

The **Exponential distribution** models the continuous time elapsed between independent random events occurring at a constant average rate $\lambda > 0$ (such as service times, component lifetimes, and inter-arrival times in queueing systems).

Definition: Exponential Distribution Exp(□)

A continuous random variable X has an **Exponential distribution** with rate parameter $\lambda > 0$ ($X \sim \text{Exp}(\lambda)$) if its PDF is:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{92}$$

3.1 Derivation of CDF, Mean, Variance and MGF

1. **Cumulative Distribution Function (CDF):** For $x \geq 0$:

$$F_X(x) = \int_0^x \lambda e^{-\lambda t}\, dt = \left[-e^{-\lambda t}\right]_0^x = 1 - e^{-\lambda x} \tag{93}$$

The survival (tail) probability is:

$$P(X > x) = 1 - F_X(x) = e^{-\lambda x} \tag{94}$$

2. **Moment Generating Function (MGF):** For $t < \lambda$:

$$M_X(t) = \int_0^\infty e^{tx} \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty e^{-(\lambda - t)x}\, dx = \frac{\lambda}{\lambda - t} = \left(1 - \frac{t}{\lambda}\right)^{-1} \tag{95}$$

Expanding in powers of t/λ:

$$M_X(t) = 1 + \frac{t}{\lambda} + \frac{t^2}{\lambda^2} + \frac{t^3}{\lambda^3} + \cdots = 1 + \left(\frac{1}{\lambda}\right) t + \left(\frac{2}{\lambda^2}\right) \frac{t^2}{2!} + \cdots \tag{96}$$

3. **Mean and Variance:**

$$\mu = E[X] = \frac{1}{\lambda}, \qquad E[X^2] = \frac{2}{\lambda^2} \implies \text{Var}(X) = E[X^2] - (E[X])^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} \tag{97}$$

Standard deviation is $\sigma = \frac{1}{\lambda}$. Notice that for the Exponential distribution: $\sigma = \mu$.

3.2 The Memoryless Property (Markov Property)

Theorem: Memoryless Property of the Exponential Distribution

Let $X \sim \text{Exp}(\lambda)$. For any non-negative time increments $s, t \geq 0$:

$$P(X > s + t \mid X > s) = P(X > t) \tag{98}$$

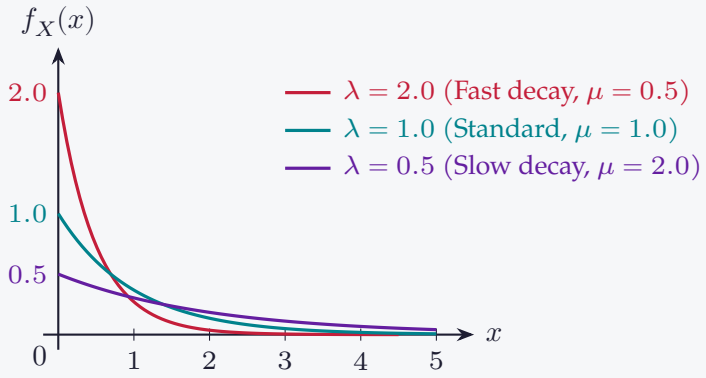
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Proof: Using the conditional probability definition and $P(X > x) = e^{-\lambda x}$:

$$P(X > s+t \mid X > s) = \frac{P(\{X > s+t\} \cap \{X > s\})}{P(X > s)}$$
$$= \frac{P(X > s+t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = \frac{e^{-\lambda s}e^{-\lambda t}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t) \tag{99}$$

Physical Interpretation: If a semiconductor component has survived s hours without failing, the probability that it survives an additional t hours is identical to the probability that a brand-new component survives t hours. The component suffers no wear-out aging over time!

TikZ Plot: Exponential Probability Density Functions for Different Rates λ



4. The Gamma Distribution and Its Properties

The **Gamma distribution** is a versatile two-parameter continuous distribution that extends the exponential distribution to model the waiting time until the occurrence of α successive Poisson events, as well as lifetime data exhibiting wear-out characteristics.

Definition: The Gamma Function $\Gamma(\alpha)$

The complete **Eulerian Gamma Function** is defined for all $\alpha > 0$ by the improper definite integral:

$$\Gamma(\alpha) = \int_0^\infty u^{\alpha-1}e^{-u}\,du \tag{100}$$

Fundamental Properties of $\Gamma(\alpha)$:

1. **Recurrence Relation:** $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ (integrating by parts).
2. **Integer Factorial:** For any positive integer $n \in \mathbb{N}$: $\Gamma(n) = (n-1)!$ with $\Gamma(1) = 0! = 1$.
3. **Half-Integer Values:** $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, from which $\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}$, $\Gamma\left(\frac{5}{2}\right) = \frac{3}{4}\sqrt{\pi}$.

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Two-Parameter Gamma Density Gamma(□, □)

A random variable X follows a **Gamma distribution** with shape parameter $\alpha > 0$ and rate parameter $\beta > 0$ (scale parameter $\theta = 1/\beta$) if its PDF is:

$$f_X(x) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \tag{101}$$

4.1 Derivation of Mean, Variance and MGF of Gamma Distribution

Rigorous Derivation: Moments of Gamma Distribution

Using the MGF definition for $t < \beta$:

$$M_X(t) = \int_0^\infty e^{tx} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}\, dx = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\beta-t)x}\, dx \tag{102}$$

Substitute $u = (\beta - t)x \implies x = \frac{u}{\beta - t}$ and $dx = \frac{du}{\beta - t}$:

$$M_X(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty \left(\frac{u}{\beta - t}\right)^{\alpha-1} e^{-u} \frac{du}{\beta - t} = \frac{\beta^\alpha}{(\beta - t)^\alpha} \cdot \underbrace{\frac{1}{\Gamma(\alpha)} \int_0^\infty u^{\alpha-1} e^{-u} du}_{=1} = \left(1 - \frac{t}{\beta}\right)^{-\alpha} \tag{103}$$

Expanding $(1 - t/\beta)^{-\alpha}$ using the generalized binomial theorem:

$$\begin{aligned} M_X(t) &= 1 + (-\alpha)\left(-\frac{t}{\beta}\right) + \frac{(-\alpha)(-\alpha-1)}{2!}\left(-\frac{t}{\beta}\right)^2 + \cdots \\ &= 1 + \left(\frac{\alpha}{\beta}\right) t + \frac{\alpha(\alpha+1)}{\beta^2} \frac{t^2}{2!} + \cdots \end{aligned} \tag{104}$$

Equating coefficients of t and $t^2/2!$:

$$E[X] = \mu = \frac{\alpha}{\beta} \tag{105}$$

$$E[X^2] = \frac{\alpha(\alpha+1)}{\beta^2} \implies \text{Var}(X) = E[X^2] - (E[X])^2 = \frac{\alpha^2 + \alpha}{\beta^2} - \frac{\alpha^2}{\beta^2} = \frac{\alpha}{\beta^2} \tag{106}$$

4.2 Special Cases and Reproductive Properties of Gamma Distribution

1. **Exponential Distribution Special Case:** When shape parameter $\alpha = 1$:

$$f_X(x) = \frac{\beta^1}{\Gamma(1)} x^{1-1} e^{-\beta x} = \beta e^{-\beta x} \equiv \text{Exp}(\beta) \tag{107}$$

2. **Erlang Distribution:** When shape parameter $\alpha = k$ is a positive integer ($k \in \mathbb{N}$), the distribution represents the sum of k independent and identically distributed $\text{Exp}(\beta)$ variables.

3. **Chi-Square (χ^2) Distribution Connection:** Substituting $\alpha = \frac{n}{2}$ and $\beta = \frac{1}{2}$ into the

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Gamma PDF yields:

$$f_X(x) = \frac{(1/2)^{n/2}}{\Gamma(n/2)} x^{n/2-1} e^{-x/2} \equiv \chi^2(n) \quad \text{(Chi-square with } n \text{ degrees of freedom)} \tag{108}$$

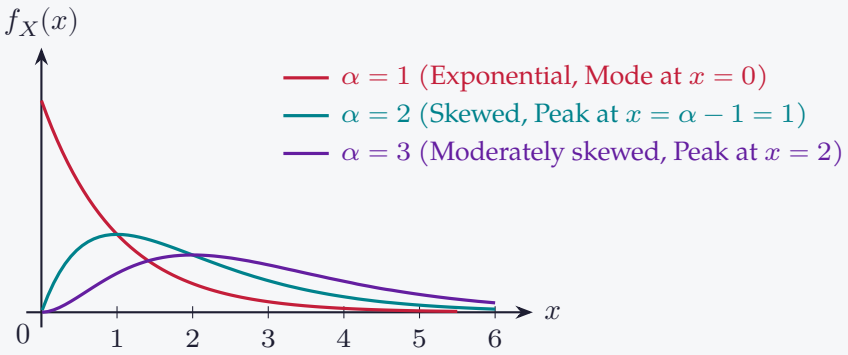
4. **Reproductive (Additive) Property:** If $X_1 \sim \text{Gamma}(\alpha_1, \beta)$ and $X_2 \sim \text{Gamma}(\alpha_2, \beta)$ are independent random variables sharing the *same* rate parameter β, then:

$$M_{X_1+X_2}(t) = M_{X_1}(t) \cdot M_{X_2}(t) = \left(1 - \frac{t}{\beta}\right)^{-\alpha_1} \cdot \left(1 - \frac{t}{\beta}\right)^{-\alpha_2} = \left(1 - \frac{t}{\beta}\right)^{-(\alpha_1+\alpha_2)} \tag{109}$$

By the uniqueness theorem of MGFs:

$$X_1 + X_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta) \tag{110}$$

TikZ Plot: Gamma Family PDFs for $\beta = 1.0$ and Varying Shape Parameter α



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Quick Revision — Module 2

Continuous Probability Distributions | BS-M301

Key Formulas and Continuous Distribution Densities at a Glance

Continuous PDF Conditions	$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x)dx = 1, \quad P(a \leq X \leq b) = \int_a^b f_X(x)dx$
CDF to PDF Relationship	$F_X(x) = \int_{-\infty}^{x} f_X(t)dt, \quad f_X(x) = \frac{d}{dx}F_X(x), \quad P(X = c) = 0$
Continuous Expectation	$E[X] = \mu = \int_{-\infty}^{\infty} x f_X(x)dx, \quad \text{Var}(X) = E[X^2] - \mu^2$
MGF Formula	$M_X(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f_X(x)dx, \quad \mu_r' = \frac{d^r M_X(t)}{dt^r}\Big\|_{t=0}$
Normal PDF $\mathcal{N}(\mu, \sigma^2)$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$
Standard Normal Z	$Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1), \quad \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(-z) = 1 - \Phi(z)$
Normal Moments	All odd $\mu_{2k+1} = 0, \quad \mu_2 = \sigma^2, \quad \mu_4 = 3\sigma^4, \quad \beta_1 = 0, \ \beta_2 = 3$
Exponential PDF Exp(λ)	$f(x) = \lambda e^{-\lambda x} \ (x \geq 0), \quad F(x) = 1 - e^{-\lambda x}, \quad \mu = \frac{1}{\lambda}, \ \sigma^2 = \frac{1}{\lambda^2}$
Memoryless Property	$P(X > s + t \mid X > s) = P(X > t)$ (Unique to Exponential)
Gamma Function	$\Gamma(\alpha) = \int_0^{\infty} u^{\alpha-1} e^{-u} du, \quad \Gamma(\alpha+1) = \alpha\Gamma(\alpha), \quad \Gamma(1/2) = \sqrt{\pi}$
Gamma PDF Gamma(α, β)	$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad \mu = \frac{\alpha}{\beta}, \quad \sigma^2 = \frac{\alpha}{\beta^2}, \quad M_X(t) = \left(1 - \frac{t}{\beta}\right)^{-\alpha}$

One-Line Definitions

- **Continuous Random Variable:** Variable taking any real value in an interval, with single-point probability $P(X = c) = 0$.
- **Probability Density Function (PDF):** Non-negative derivative of the CDF whose integral over an interval gives probability.
- **Normal Distribution:** Symmetrical mesokurtic bell curve fully governed by location μ and scale σ.
- **Inflection Points:** Points where curvature flips concavity, situated at precisely $x = \mu \pm \sigma$ for a normal curve.
- **Standard Score (Z):** Dimensionless measure $Z = (X - \mu)/\sigma$ expressing deviation from mean in units of σ.
- **Exponential Distribution:** Only continuous distribution possessing memorylessness, modelling Poisson inter-arrival times.
- **Memoryless Property:** Independence of future remaining lifetime from past survival duration: $P(X > s + t \mid X > s) = P(X > t)$.
- **Gamma Function ($\Gamma(\alpha)$):** Generalized factorial integral extending $n!$ to all real positive numbers ($\Gamma(n) = (n-1)!$).
- **Erlang Distribution:** Gamma distribution with integer shape parameter $\alpha = k$, representing sum of k independent exponentials.
- **Reproductive Property:** Closure under convolution: sum of independent Gammas with same β yields another Gamma.

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Frequently Asked / Viva Questions (Part 1)

Q1. Why is the probability of a continuous random variable taking an exact value zero ($P(X = c) = 0$)?

By definition, $P(X = c) = \int_c^c f_X(x)\,dx = 0$. Since an interval contains uncountably infinitely many points, the probability mass allocated to any isolated geometric point vanishes, even though the event is not impossible in reality.

Q2. Can a probability density function exceed unity ($f_X(x) > 1$)? Explain with an example.

Yes! Unlike discrete PMFs where $p(x) \leq 1$, a continuous PDF represents probability *density* per unit width, not probability. For example, for $X \sim U[0, 0.2]$, $f_X(x) = 1/(0.2 - 0) = 5.0 > 1$, while the total integral $\int_0^{0.2} 5.0\,dx = 1.0$ is fully preserved.

Q3. Derive the Moment Generating Function of the standard normal distribution $Z \sim \mathcal{N}(0, 1)$.

$M_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz} e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z^2-2tz)/2} dz$. Completing the square: $z^2 - 2tz = (z - t)^2 - t^2$. Factoring out $e^{t^2/2}$: $M_Z(t) = e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-t)^2/2} dz = e^{t^2/2} \cdot 1 = e^{t^2/2}$.

Q4. Prove that all odd central moments of a Normal distribution are identically zero.

$\mu_{2k+1} = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^{2k+1} e^{-(x-\mu)^2/(2\sigma^2)} dx$. Substituting $u = x - \mu$: the integrand $u^{2k+1} e^{-u^2/(2\sigma^2)}$ is an odd function of u ($g(-u) = -g(u)$) integrated over symmetric limits $(-\infty, \infty)$, which integrates to zero.

Q5. Where do the inflection points of a normal probability curve occur?

Equating the second derivative to zero: $f''(x) = \left[\frac{(x-\mu)^2}{\sigma^4} - \frac{1}{\sigma^2}\right] f(x) = 0 \implies (x - \mu)^2 = \sigma^2 \implies x = \mu \pm \sigma$. The curve transitions from concave downwards to concave upwards at exactly one standard deviation on either side of the mean.

Q6. State and prove the memoryless property of the Exponential distribution.

$P(X > s + t \mid X > s) = \frac{P(X>s+t)}{P(X>s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$. The conditional probability that a component lasts t more units given it has lasted s units is independent of s.

Q7. What is the physical connection between the Poisson process and the Exponential distribution?

In a Poisson process with rate λ, the number of event arrivals in interval t follows Poisson(λt). The waiting time T until the first arrival exceeds t if and only if zero arrivals occur in $[0, t]$: $P(T > t) = P(N(t) = 0) = \frac{(\lambda t)^0 e^{-\lambda t}}{0!} = e^{-\lambda t}$. Thus, the inter-arrival time T is exponentially distributed with parameter λ.

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Frequently Asked / Viva Questions (Part 2)

Q8. Prove that $\Gamma(1/2) = \sqrt{\pi}$.

$\Gamma(1/2) = \int_0^\infty u^{-1/2} e^{-u} du$. Substituting $u = z^2/2 \implies du = zdz$: $\Gamma(1/2) = \int_0^\infty \frac{\sqrt{2}}{z} e^{-z^2/2} z dz = \sqrt{2}\int_0^\infty e^{-z^2/2} dz$. By symmetry: $\int_0^\infty e^{-z^2/2} dz = \frac{1}{2}\int_{-\infty}^\infty e^{-z^2/2} dz = \frac{\sqrt{2\pi}}{2} = \sqrt{\frac{\pi}{2}}$. Multiplying by $\sqrt{2}$ yields $\Gamma(1/2) = \sqrt{\pi}$.

Q9. How is the Chi-Square (χ^2) distribution obtained as a special case of the Gamma distribution?

Setting shape parameter $\alpha = n/2$ and rate parameter $\beta = 1/2$ in the Gamma PDF yields $f(x) = \frac{(1/2)^{n/2}}{\Gamma(n/2)} x^{n/2-1} e^{-x/2}$, which is the standard definition of the Chi-Square distribution with n degrees of freedom.

Q10. State the reproductive property of independent Normal random variables.

If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, any linear combination $Y = aX_1 + bX_2$ is also normally distributed: $Y \sim \mathcal{N}(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$. This follows directly from multiplying their MGFs.

Summary Matrix: Comprehensive Comparison of Continuous Distributions

Distribution	Probability Density Function (PDF)	Mean ($E[X]$)	Variance (σ^2)	Key Engineering Application
Uniform $U[a,b]$	$f(x) = \frac{1}{b-a}, \quad x \in [a,b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	Quantization noise in ADC systems.
Normal $\mathcal{N}(\mu, \sigma^2)$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2	Thermal Gaussian noise; Central Limit Theorem.
Exponential **Exp**(λ)	$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	Lifetime of electronic parts; queueing theory.
Gamma (α, β)	$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \ x > 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	Waiting time for α events; reliability engineering.
Chi-Square $\chi^2(n)$	$f(x) = \frac{(1/2)^{n/2}}{\Gamma(n/2)} x^{n/2-1} e^{-x/2}$	n	$2n$	Goodness-of-fit testing; sample variance distribution.

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Module 3

Bivariate Distributions and Their Fundamental Properties

1. Bivariate Distributions and Their Fundamental Properties

In numerous engineering problems, physical phenomena involve the simultaneous observation of two interdependent random quantities, such as the in-phase and quadrature components of an RF communication signal (I, Q), transmitter voltage and received current (V, I), or temperature and humidity. These are modeled jointly as a **two-dimensional (bivariate) random variable** $(X, Y) : \Omega \to \mathbb{R}^2$.

Definition: Joint Probability Mass and Density Functions

Discrete Bivariate Case (Joint PMF): Let (X, Y) take values in a countable set of pairs (x_i, y_j). The **Joint PMF** is:

$$p_{XY}(x_i, y_j) = P(X = x_i, Y = y_j) \tag{111}$$

Axioms: (1) $p_{XY}(x_i, y_j) \geq 0, \quad \forall (i, j), \quad$ (2) $\sum_i \sum_j p_{XY}(x_i, y_j) = 1.$

Continuous Bivariate Case (Joint PDF): (X, Y) is jointly continuous if there exists an integrable surface $f_{XY} : \mathbb{R}^2 \to \mathbb{R}$ such that for any two-dimensional region $D \subseteq \mathbb{R}^2$:

$$P((X, Y) \in D) = \iint_D f_{XY}(x, y)\, dxdy \tag{112}$$

Axioms:

$$\textbf{Non-negativity:} \quad f_{XY}(x, y) \geq 0, \quad \forall (x, y) \in \mathbb{R}^2 \tag{113}$$

$$\textbf{Unit Volume:} \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y)\, dxdy = 1 \tag{114}$$

1.1 Joint Cumulative Distribution Function (Joint CDF)

The **Joint CDF** of (X, Y) is defined as the probability of the infinite quadrant to the south-west of (x, y):

$$F_{XY}(x, y) = P(X \leq x, Y \leq y) = \begin{cases} \sum_{x_i \leq x} \sum_{y_j \leq y} p_{XY}(x_i, y_j), & \text{(Discrete)} \\ \int_{-\infty}^{x} \int_{-\infty}^{y} f_{XY}(u, v)\, dvdu, & \text{(Continuous)} \end{cases} \tag{115}$$

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Properties of the Joint CDF and Recovery of Joint PDF

1. **Boundary Limits:**

$$F_{XY}(-\infty, y) = 0, \qquad F_{XY}(x, -\infty) = 0, \qquad F_{XY}(-\infty, -\infty) = 0 \tag{116}$$

$$F_{XY}(+\infty, +\infty) = 1 \tag{117}$$

2. **Marginal Recovery from Joint CDF:**

$$F_X(x) = \lim_{y \to +\infty} F_{XY}(x, y) = F_{XY}(x, \infty), \qquad F_Y(y) = \lim_{x \to +\infty} F_{XY}(x, y) = F_{XY}(\infty, y) \tag{118}$$

3. **Recovery of Continuous Joint PDF by Partial Differentiation:** At every continuity point of $f_{XY}(x, y)$:

$$f_{XY}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x, y) = \frac{\partial^2}{\partial y \partial x} F_{XY}(x, y) \tag{119}$$

4. **Rectangle Probability Rule:** For any rectangular region $[x_1, x_2] \times [y_1, y_2]$:

$$P(x_1 < X \le x_2,\ y_1 < Y \le y_2) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1) \ge 0 \tag{120}$$

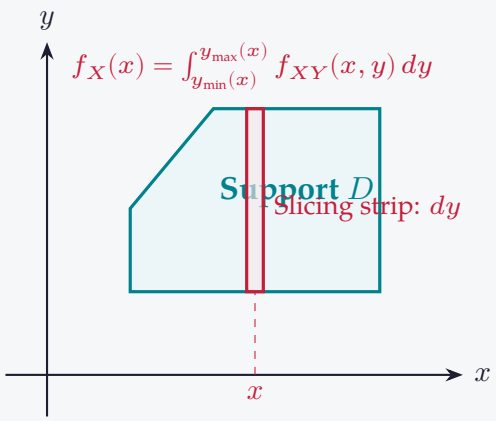
1.2 Marginal Distributions

The individual distribution of one variable obtained by integrating or summing out the other variable over its entire support is termed the **marginal distribution**.

$$\textbf{Marginal PMFs:} \quad p_X(x_i) = \sum_j p_{XY}(x_i, y_j), \qquad p_Y(y_j) = \sum_i p_{XY}(x_i, y_j) \tag{121}$$

$$\textbf{Marginal PDFs:} \quad f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx \tag{122}$$

TikZ Schematic: Geometric Interpretation of Marginal Integration across Joint Support



Area under the cross-sectional slice at fixed x gives the marginal density $f_X(x)$.

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2. Conditional Distributions, Densities and Bayes' Rule

Definition: Conditional Probability Density Function

Given that the random variable Y has taken on a specific numerical value $Y = y$ (where marginal density $f_Y(y) > 0$), the **conditional PDF** of X given $Y = y$ is defined as:

$$f_{X|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{f_{XY}(x,y)}{\displaystyle\int_{-\infty}^{\infty} f_{XY}(u,y)\,du} \tag{123}$$

Symmetrically, the conditional PDF of Y given $X = x$ (with $f_X(x) > 0$) is:

$$f_{Y|X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{f_{XY}(x,y)}{\displaystyle\int_{-\infty}^{\infty} f_{XY}(x,v)\,dv} \tag{124}$$

Every conditional density satisfies the standard PDF axioms: $f_{X|Y}(x \mid y) \geq 0$ and $\int_{-\infty}^{\infty} f_{X|Y}(x \mid y)dx = 1$.

2.1 Conditional Expectation and the Tower Property

- **Conditional Expectation:** The mean of X given $Y = y$ is a function of y:

$$E[X \mid Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x \mid y)\,dx \tag{125}$$

- **Law of Iterated Expectations (Tower Property):**

$$E\left[E[X \mid Y]\right] = E[X] \tag{126}$$

Proof:

$$\begin{aligned} E\left[E[X \mid Y]\right] &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x \frac{f_{XY}(x,y)}{f_Y(y)} dx \right) f_Y(y)\,dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x,y)\,dxdy = \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{XY}(x,y)dy \right) dx \\ &= \int_{-\infty}^{\infty} x f_X(x)\,dx = E[X] \end{aligned} \tag{127}$$

- **Law of Total Variance (Eve's Law):**

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y]) \tag{128}$$

Total uncertainty equals the average unexplained variance plus the variance explained by Y.

2.2 Bayes' Rule for Continuous and Mixed Random Variables

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Continuous and Mixed Formulations of Bayes' Rule

Continuous Inversion (Posterior Density): Equating $f_{XY}(x,y) = f_X(x)f_{Y|X}(y \mid x) = f_Y(y)f_{X|Y}(x \mid y)$:

$$f_{X|Y}(x \mid y) = \frac{f_X(x) \cdot f_{Y|X}(y \mid x)}{f_Y(y)} = \frac{f_X(x) \cdot f_{Y|X}(y \mid x)}{\displaystyle\int_{-\infty}^{\infty} f_X(u) \cdot f_{Y|X}(y \mid u)\, du} \tag{129}$$

where $f_X(x)$ is the *prior density*, $f_{Y|X}(y \mid x)$ is the *likelihood function*, and $f_{X|Y}(x \mid y)$ is the *posterior density*.

Discrete Hypothesis with Continuous Measurement (Signal Detection): Let $\{H_1, H_2, \ldots, H_n\}$ be discrete transmitted symbols (priors $P(H_i)$) observed through continuous additive noise Y:

$$P(H_k \mid Y = y) = \frac{P(H_k) \cdot f_{Y|H_k}(y)}{\displaystyle\sum_{i=1}^{n} P(H_i) \cdot f_{Y|H_i}(y)} \tag{130}$$

3. Bivariate Transformations and the Jacobian Method

When analyzing signal processing systems, inputs (X, Y) are mapped into new output variables (U, V) via mathematical functions $U = g_1(X, Y)$ and $V = g_2(X, Y)$.

Theorem: Bivariate Jacobian Transformation

Let (X, Y) have joint PDF $f_{XY}(x, y)$ supported on set D. Let the transformation:

$$u = g_1(x, y), \qquad v = g_2(x, y) \tag{131}$$

define a one-to-one and invertible mapping from D onto range S, with unique inverse:

$$x = h_1(u, v), \qquad y = h_2(u, v) \tag{132}$$

If the partial derivatives are continuous, the **Jacobian determinant** J of the inverse transformation is:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \tag{133}$$

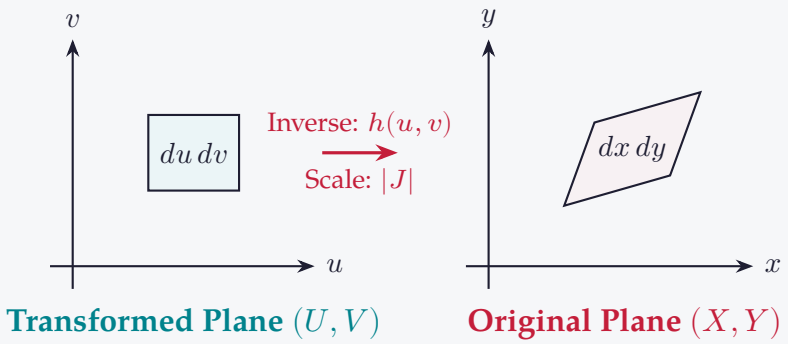
Then the joint PDF of (U, V) over S is given by:

$$f_{UV}(u, v) = f_{XY}\left(h_1(u, v), h_2(u, v)\right) \cdot |J| \tag{134}$$

where $|J|$ is the absolute value of the Jacobian determinant (representing the local area scaling factor: $dxdy = |J|dudv$).

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TikZ Schematic: Area Element Transformation under Jacobian Mapping



4. Distributions of Sums and Quotients

4.1 Distribution of the Sum $Z = X + Y$ (Convolution)

Derivation: PDF of the Sum of Two Continuous Random Variables

To find the PDF of $Z = X + Y$, introduce an auxiliary variable $W = X$. The forward transformation is:

$$z = x + y, \qquad w = x \tag{135}$$

The unique inverse transformation is:

$$x = w, \qquad y = z - w \tag{136}$$

Computing the Jacobian determinant:

$$J = \det \begin{pmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{pmatrix} = \det \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} = (0)(-1) - (1)(1) = -1 \implies |J| = |-1| = 1 \tag{137}$$

The joint PDF of (Z, W) is:

$$f_{ZW}(z, w) = f_{XY}(w, z - w) \cdot |J| = f_{XY}(w, z - w) \tag{138}$$

Integrating out the auxiliary variable w yields the marginal PDF of the sum Z:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{ZW}(z, w)\, dw = \int_{-\infty}^{\infty} f_{XY}(w, z - w)\, dw \tag{139}$$

If X and Y are **independent** ($f_{XY}(x, y) = f_X(x) f_Y(y)$), this becomes the fundamental **continuous convolution integral**:

$$f_Z(z) = (f_X * f_Y)(z) = \int_{-\infty}^{\infty} f_X(w) f_Y(z - w)\, dw = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y)\, dy \tag{140}$$

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4.2 Distribution of the Quotient $Z = X/Y$

Derivation: PDF of the Quotient of Two Random Variables

Let $Z = \frac{X}{Y}$. Introduce the auxiliary variable $W = Y$. The forward transformation is:

$$z = \frac{x}{y}, \qquad w = y \quad (w \neq 0) \tag{141}$$

The inverse transformation is:

$$x = z \cdot w, \qquad y = w \tag{142}$$

The Jacobian determinant is:

$$J = \det \begin{pmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{pmatrix} = \det \begin{pmatrix} w & z \\ 0 & 1 \end{pmatrix} = (w)(1) - (z)(0) = w \implies |J| = |w| \tag{143}$$

The joint PDF of (Z, W) is:

$$f_{ZW}(z, w) = f_{XY}(zw, w) \cdot |w| \tag{144}$$

Integrating out w over $(-\infty, \infty)$ yields the general quotient density:

$$f_Z(z) = \int_{-\infty}^{\infty} |w| f_{XY}(zw, w)\, dw \tag{145}$$

If X and Y are independent:

$$f_Z(z) = \int_{-\infty}^{\infty} |w| f_X(zw) f_Y(w)\, dw \tag{146}$$

4.3 Quotient of Two Independent Standard Normals: The Cauchy Distribution

Classic Result: Ratio of Independent Standard Normals follows Cauchy

Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 1)$ be independent. Their joint density is:

$$f_{XY}(x, y) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2} = \frac{1}{2\pi} e^{-(x^2+y^2)/2} \tag{147}$$

Applying the quotient formula to $Z = X/Y$:

$$f_Z(z) = \int_{-\infty}^{\infty} |w| \cdot \frac{1}{2\pi} e^{-(z^2 w^2 + w^2)/2}\, dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} |w| e^{-\frac{w^2(1+z^2)}{2}}\, dw \tag{148}$$

Since the integrand is even with respect to w:

$$f_Z(z) = \frac{2}{2\pi} \int_0^{\infty} w e^{-\frac{w^2(1+z^2)}{2}}\, dw = \frac{1}{\pi} \int_0^{\infty} w e^{-\frac{w^2(1+z^2)}{2}}\, dw \tag{149}$$

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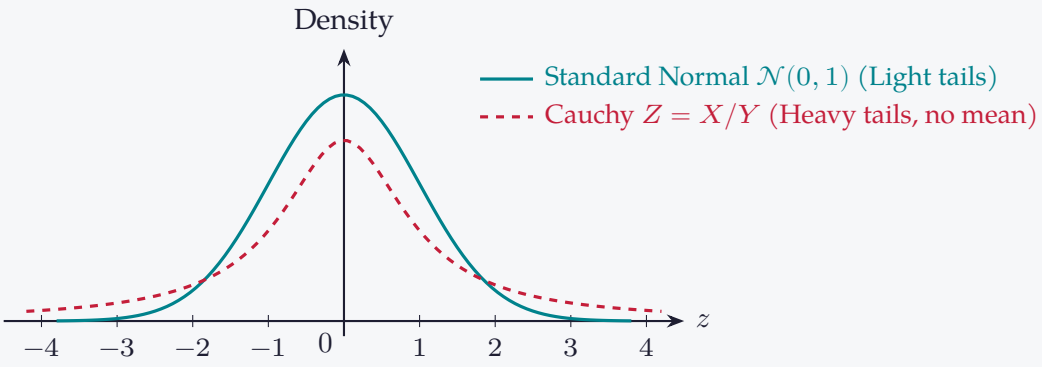
Substitute $u = \frac{w^2(1+z^2)}{2} \implies du = (1+z^2)wdw \implies wdw = \frac{du}{1+z^2}$:

$$f_Z(z) = \frac{1}{\pi(1+z^2)} \int_0^\infty e^{-u}\, du = \frac{1}{\pi(1+z^2)} \left[-e^{-u}\right]_0^\infty = \frac{1}{\pi(1+z^2)}(1) \tag{150}$$

$$f_Z(z) = \frac{1}{\pi(1+z^2)}, \qquad -\infty < z < \infty \tag{151}$$

This is the celebrated **Standard Cauchy Distribution**! *Crucial Exam Note:* For a Cauchy distribution, $\int_{-\infty}^{\infty} |z| \frac{1}{\pi(1+z^2)} dz = \infty$. Therefore, the Cauchy distribution has **no defined mean, no variance, and no MGF**!

TikZ Plot: Standard Cauchy PDF vs. Standard Normal Gaussian PDF



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Quick Revision — Module 3

Bivariate Distributions, Sums, Quotients and Bayes' Rule | BS-M301

Key Formulas and Bivariate Relations at a Glance

Continuous Joint PDF	$f_{XY}(x,y) \geq 0, \quad \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,y)dxdy = 1$
Joint CDF Formula	$F_{XY}(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{XY}(u,v)dvdu, \quad f_{XY}(x,y) = \frac{\partial^2 F_{XY}}{\partial x \partial y}$
Marginal Densities	$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)dx$
Conditional Densities	$f_{X\|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)}, \quad f_{Y\|X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)}$
Statistical Independence	$f_{XY}(x,y) = f_X(x)f_Y(y) \iff f_{X\|Y}(x \mid y) = f_X(x)$
Tower Property	$E\left[E[X \mid Y]\right] = E[X], \quad \text{Var}(X) = E[\text{Var}(X\|Y)] + \text{Var}(E[X\|Y])$
Jacobian Transform	$f_{UV}(u,v) = f_{XY}(h_1,h_2)\cdot\|J\|, \quad J = \det\begin{pmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{pmatrix}$
Sum PDF (Convolution)	$Z = X+Y \implies f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(w, z-w)dw$
Quotient PDF	$Z = X/Y \implies f_Z(z) = \int_{-\infty}^{\infty} \|w\| f_{XY}(zw, w)dw$
Cauchy from Normals	$X, Y \sim \mathcal{N}(0,1)$ indep. $\implies Z = \frac{X}{Y} \sim$ Cauchy : $f_Z(z) = \frac{1}{\pi(1+z^2)}$
Continuous Bayes' Rule	$f_{X\|Y}(x \mid y) = \frac{f_X(x)f_{Y\|X}(y\|x)}{\int_{-\infty}^{\infty} f_X(u)f_{Y\|X}(y\|u)du}$

One-Line Definitions

- **Joint PDF $f_{XY}(x,y)$:** Surface whose double integral over any two-dimensional region D gives $P((X,Y) \in D)$.
- **Marginal PDF:** Individual density of one component obtained by integrating out the other variable over $\mathbb{R}$.
- **Conditional PDF:** Sliced cross-sectional density of X normalized by $f_Y(y)$ under the condition that $Y = y$.
- **Statistical Independence:** Complete factorization $f_{XY}(x,y) = f_X(x)f_Y(y)$ across the entire domain.
- **Tower Property:** Fundamental law asserting that the overall expectation equals the expectation of conditional expectations.
- **Jacobian Determinant (J):** Local area transformation ratio $dxdy = |J|dudv$ relating transformed coordinate spaces.
- **Convolution Theorem:** Operation yielding the PDF of the sum of independent random variables: $f_Z = f_X * f_Y$.
- **Quotient Distribution:** Density $f_Z(z) = \int |w| f(zw, w)dw$ describing the ratio of two random variables.
- **Cauchy Distribution:** Heavy-tailed distribution $f(z) = \frac{1}{\pi(1+z^2)}$ resulting from $\mathcal{N}(0,1)/\mathcal{N}(0,1)$, lacking mean and variance.
- **Posterior Density:** Updated probability density of an unknown parameter after incorporating experimental observation y.

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Frequently Asked / Viva Questions (Part 1)

Q1. State the necessary and sufficient conditions for two continuous random variables X and Y to be independent.

X and Y are independent if and only if their joint PDF factors into the product of their marginal PDFs everywhere: $f_{XY}(x,y) = f_X(x)f_Y(y)$ for all $(x,y) \in \mathbb{R}^2$, and the support region D is a product of intervals (non-coupled boundaries).

Q2. How is the joint PDF obtained from the joint Cumulative Distribution Function?

By mixed second-order partial differentiation: $f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x,y)$, provided the second partial derivatives exist and are continuous.

Q3. State and prove the Law of Iterated Expectations (Tower Property).

$E\left[E[X \mid Y]\right] = E[X]$. Proof: $E[E[X \mid Y]] = \int \left(\int x \frac{f(x,y)}{f_Y(y)} dx \right) f_Y(y) dy = \iint x f(x,y) dx dy = \int x f_X(x) dx = E[X]$.

Q4. What is the physical significance of the Jacobian determinant in bivariate transformations?

The Jacobian determinant $J = \frac{\partial(x,y)}{\partial(u,v)}$ represents the local geometric dilation factor between the original and transformed coordinate planes, ensuring conservation of probability mass: $P(dxdy) = f_{XY}(x,y)dxdy = f_{XY}(h_1,h_2)|J|dudv = f_{UV}(u,v)dudv$.

Q5. Derive the PDF of the sum $Z = X + Y$ of two independent continuous random variables.

Let $Z = X + Y, W = X \implies X = W, Y = Z - W$. The Jacobian is $|J| = 1$. The joint density is $f_{ZW}(z,w) = f_{XY}(w, z-w) = f_X(w)f_Y(z-w)$. Integrating over w gives the convolution: $f_Z(z) = \int_{-\infty}^{\infty} f_X(w)f_Y(z-w)dw$.

Q6. Why does the absolute value $|w|$ appear in the formula for the quotient density $Z = X/Y$?

The Jacobian of the inverse transformation $x = zw, y = w$ is $J = \det \begin{pmatrix} w & z \\ 0 & 1 \end{pmatrix} = w$.

Because an area element must always be strictly non-negative ($dxdy = |J|dzdw$), the absolute value $|J| = |w|$ must be taken.

Q7. Show that the ratio of two independent standard normal random variables follows a Cauchy distribution.

$f_Z(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |w| e^{-w^2(1+z^2)/2} dw = \frac{1}{\pi} \int_0^{\infty} w e^{-w^2(1+z^2)/2} dw$. Substituting $u = w^2(1+z^2)/2$ transforms the integral to $\frac{1}{\pi(1+z^2)} \int_0^{\infty} e^{-u} du = \frac{1}{\pi(1+z^2)}$, which is the standard Cauchy PDF.

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Frequently Asked / Viva Questions (Part 2)

Q8. Why does the Cauchy distribution have no mathematical expectation or variance?

$E[|Z|] = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{|z|}{1+z^2}dz = \frac{2}{\pi}\int_0^{\infty}\frac{z}{1+z^2}dz = \frac{1}{\pi}\left[\ln(1+z^2)\right]_0^{\infty} = \infty$. Because the absolute integral diverges logarithmically due to heavy polynomial tails ($\sim 1/z^2$), the mean is mathematically undefined, and all higher moments diverge.

Q9. State the Law of Total Variance (Eve's Law) and interpret its terms.

$\text{Var}(X) = E[\text{Var}(X \mid Y)] + \text{Var}(E[X \mid Y])$. The first term represents the average unexplained variation remaining in X after conditioning on Y; the second term represents the variation in the conditional predictor $E[X \mid Y]$ explained by Y.

Q10. State Continuous Bayes' Theorem and explain its role in signal estimation.

$f_{X|Y}(x \mid y) = \frac{f_X(x)f_{Y|X}(y|x)}{\int f_X(u)f_{Y|X}(y|u)du}$. In communications, X is the transmitted signal (prior), $f_{Y|X}(y \mid x)$ is the channel noise model (likelihood), and $f_{X|Y}(x \mid y)$ is the posterior distribution used by the receiver to make Maximum A Posteriori (MAP) symbol decisions.

Summary Matrix: Comprehensive Comparison of Bivariate Operations

Operation / Concept	Mathematical Governing Expression	Independence Simplification	Physical Meaning / Role
Marginal Density	$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dy$	$f_X(x)\int f_Y(y)dy = f_X(x)$	Complete projection onto X-axis.
Conditional Density	$f_{X\|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)}$	$f_{X\|Y}(x \mid y) = f_X(x)$	Updated density given knowledge $Y = y$.
Sum of RVs	$Z = X + Y \implies f_Z(z) = \int f_{XY}(w, z-w)dw$	**Convolution:** $f_Z = f_X * f_Y$	Total noise from independent sources.
Quotient of RVs	$Z = X/Y \implies f_Z(z) = \int \lvert w\rvert f_{XY}(zw, w)dw$	$\int \lvert w\rvert f_X(zw)f_Y(w)dw$	Signal-to-noise ratio; slope of vector.
Cauchy Distribution	$Z = X/Y$ **for** $X, Y \sim \mathcal{N}(0,1) \implies \frac{1}{\pi(1+z^2)}$	**Quotient of two standard Gaussians**	Heavy-tailed ratio; undefined moments.
Continuous Bayes	$f_{X\|Y}(x \mid y) = \frac{f_X(x)f_{Y\|X}(y\|x)}{\int f_X(u)f_{Y\|X}(y\|u)du}$	$f_{X\|Y}(x \mid y) = f_X(x)$	Optimal Bayesian estimation / inference.

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Module 4

Measures of Central Tendency, Moments, Skewness and Kurtosis

1. Measures of Central Tendency, Moments, Skewness and Kurtosis

Descriptive statistics provides quantitative summaries describing the central clustering, dispersion, asymmetry, and peakedness of experimental datasets.

Definition: Measures of Central Tendency

1. **Arithmetic Mean ($\bar{x}$):** The sum of all observations divided by sample size:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{n} f_i x_i, \qquad N = \sum_{i=1}^{n} f_i \tag{152}$$

Key Properties: (1) Algebraic sum of deviations is zero: $\sum f_i(x_i - \bar{x}) = 0$. (2) Sum of squared deviations is minimal about the mean: $\sum f_i(x_i - A)^2$ is minimum when $A = \bar{x}$.

2. **Median (M):** The middle value partitioning the ordered dataset into two equal halves. For grouped data with median class $[L, L + h]$:

$$M = L + \frac{\frac{N}{2} - C}{f} \cdot h \tag{153}$$

where L is lower boundary, C is cumulative frequency of preceding class, and f is frequency.

3. **Mode (Z):** The most frequently occurring value in the distribution:

$$Z = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \cdot h \tag{154}$$

Empirical Relationship for Moderately Skewed Distributions:

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean} \iff \text{Mean} - \text{Mode} \approx 3(\text{Mean} - \text{Median}) \tag{155}$$

1.1 Moments and Moment Conversion Relations

- **r-th Raw Moment about Origin (μ'_r):** $\mu'_r = \frac{1}{N}\sum f_i x_i^r$. Note: $\mu'_1 = \bar{x}$.
- **r-th Central Moment about Mean (μ_r):** $\mu_r = \frac{1}{N}\sum f_i(x_i - \bar{x})^r$.

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Conversion Relations: Central Moments from Raw Moments

Expanding $(x_i - \bar{x})^r = \sum_{k=0}^{r} \binom{r}{k} x_i^{r-k}(-\bar{x})^k$:

$$\mu_1 = 0 \quad \text{(Always zero!)} \tag{156}$$
$$\mu_2 = \mu_2' - (\mu_1')^2 = \sigma^2 \quad \text{(Variance)} \tag{157}$$
$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3 \tag{158}$$
$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4 \tag{159}$$

1.2 Measures of Skewness (Asymmetry)

Skewness quantifies the departure of a distribution curve from perfect bilateral symmetry.

Formulas for Coefficients of Skewness

1. **Karl Pearson's Coefficient of Skewness (S_k):**

$$S_k = \frac{\text{Mean} - \text{Mode}}{\sigma} = \frac{3(\text{Mean} - \text{Median})}{\sigma}, \qquad -3 \le S_k \le 3 \tag{160}$$

2. **Bowley's Quartile Coefficient of Skewness (S_b):** Based on quartiles $Q_1, Q_2(\text{Median}), Q_3$:

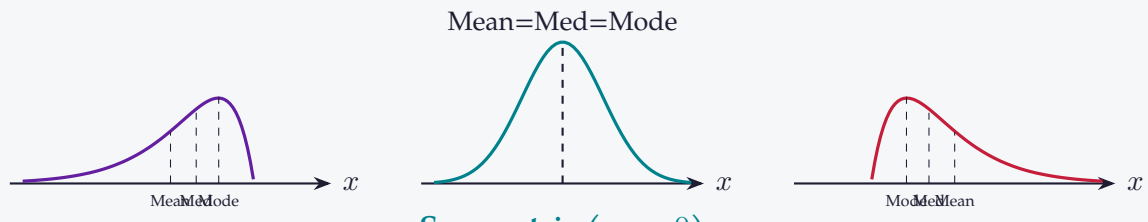
$$S_b = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)} = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}, \qquad -1 \le S_b \le 1 \tag{161}$$

3. **Moment Coefficient of Skewness (β_1 and γ_1):**

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{\mu_3^2}{\sigma^6}, \qquad \gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\sigma^3} \tag{162}$$

- $\gamma_1 = 0$: Symmetric distribution (Mean = Median = Mode).
- $\gamma_1 > 0$: Positively skewed (long tail stretched to the right; Mean > Median > Mode).
- $\gamma_1 < 0$: Negatively skewed (long tail stretched to the left; Mode > Median > Mean).

TikZ Schematic: Geometric Comparison of Symmetrical and Skewed Distributions



Negative Skew ($\gamma_1 < 0$) Symmetric ($\gamma_1 = 0$) Positive Skew ($\gamma_1 > 0$)

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1.3 Measures of Kurtosis (Peakedness)

Definition: Kurtosis Coefficients β_2 and γ_2

Kurtosis quantifies the degree of peakedness and the heaviness of the tails of a frequency curve relative to the standard normal bell curve:

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4}, \qquad \gamma_2 = \beta_2 - 3 \tag{163}$$

- **Mesokurtic** ($\beta_2 = 3, \gamma_2 = 0$)**:** Normal distribution curve profile.
- **Leptokurtic** ($\beta_2 > 3, \gamma_2 > 0$)**:** Sharply peaked center with heavy, fat tails (e.g., Student's t, Laplace).
- **Platykurtic** ($\beta_2 < 3, \gamma_2 < 0$)**:** Flatter plateau-like peak with light, rapidly vanishing tails (e.g., Uniform distribution where $\beta_2 = 1.8$).

2. Evaluation of Statistical Parameters for Core Distributions

2.1 Binomial Distribution: Parameters and Recurrence Relation

For $X \sim \text{Bin}(n, p)$ with $q = 1 - p$:

$$M_X(t) = \sum_{k=0}^{n} e^{tk} \binom{n}{k} p^k q^{n-k} = (q + pe^t)^n \tag{164}$$

$$E[X] = \mu = np, \qquad \text{Var}(X) = \sigma^2 = npq \tag{165}$$

$$\mu_3 = npq(q - p), \qquad \mu_4 = 3n^2p^2q^2 + npq(1 - 6pq) \tag{166}$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(q-p)^2}{npq}, \qquad \beta_2 = \frac{\mu_4}{\mu_2^2} = 3 + \frac{1 - 6pq}{npq} \tag{167}$$

Recurrence Relation for Central Moments:

$$\mu_{r+1} = pq \left[nr\mu_{r-1} + \frac{d\mu_r}{dp} \right] \tag{168}$$

2.2 Poisson Distribution: Parameters and Recurrence Relation

For $X \sim \text{Poisson}(\lambda)$:

$$M_X(t) = \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = \exp\left(\lambda(e^t - 1)\right) \tag{169}$$

$$E[X] = \lambda, \qquad \text{Var}(X) = \lambda \tag{170}$$

$$\mu_3 = \lambda, \qquad \mu_4 = 3\lambda^2 + \lambda \tag{171}$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{\lambda^2}{\lambda^3} = \frac{1}{\lambda}, \qquad \beta_2 = \frac{3\lambda^2 + \lambda}{\lambda^2} = 3 + \frac{1}{\lambda} \tag{172}$$

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Recurrence Relation for Central Moments:

$$\mu_{r+1} = \lambda \left[r\mu_{r-1} + \frac{d\mu_r}{d\lambda} \right] \tag{173}$$

As $\lambda \to \infty$, both $\beta_1 \to 0$ and $\beta_2 \to 3$, confirming convergence of Poisson to Normal!

2.3 Normal Distribution: Parameters and Recurrence Relation

For $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$E[X] = \mu, \qquad \text{Var}(X) = \sigma^2 \tag{174}$$

$$\mu_{2k+1} = 0 \quad \forall k \in \mathbb{N}_0, \qquad \mu_{2k} = \frac{(2k)!}{2^k k!}\sigma^{2k} \tag{175}$$

$$\beta_1 = 0, \qquad \beta_2 = \frac{3\sigma^4}{\sigma^4} = 3 \iff \gamma_2 = 0 \tag{176}$$

Recurrence Relation for Even Central Moments:

$$\mu_{2k} = (2k-1)\sigma^2 \mu_{2k-2} \tag{177}$$

3. Correlation and Regression Analysis

3.1 Karl Pearson's Coefficient of Correlation (r)

Karl Pearson's product-moment correlation coefficient r provides a scale-invariant measure of linear association between two paired variables (x_i, y_i):

Formula for Pearson's Correlation Coefficient

$$r = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} = \frac{n\sum x_i y_i - (\sum x_i)(\sum y_i)}{\sqrt{n\sum x_i^2 - (\sum x_i)^2}\sqrt{n\sum y_i^2 - (\sum y_i)^2}} \tag{178}$$

Properties of r:

1. Normalized limits: $-1 \leq r \leq +1$.
2. Invariant to change of origin and scale: If $u = \frac{x-a}{h}$ and $v = \frac{y-b}{k}$ with $h, k > 0$, then $r_{xy} = r_{uv}$.

3.2 Linear Regression Equations and Least Squares

Regression expresses the average functional relationship predicting a dependent variable from an independent variable.

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The Two Lines of Regression

1. Line of Regression of Y on X: Predicts Y for a known value of X:

$$Y - \bar{y} = b_{yx}(X - \bar{x}), \qquad b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{\mathrm{Cov}(X,Y)}{\sigma_x^2} \tag{179}$$

2. Line of Regression of X on Y: Predicts X for a known value of Y:

$$X - \bar{x} = b_{xy}(Y - \bar{y}), \qquad b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_y^2} \tag{180}$$

where b_{yx} and b_{xy} are the **regression coefficients**.

Key Properties of Regression Coefficients

1. **Geometric Mean Property:** The correlation coefficient is the geometric mean of the regression coefficients:
$$r = \pm\sqrt{b_{yx} \cdot b_{xy}} \tag{181}$$
Both regression coefficients must share the same algebraic sign, which is also the sign of r.
2. **Reciprocal Magnitude Constraint:** If $b_{yx} > 1$, then $b_{xy} < 1$, since $b_{yx}b_{xy} = r^2 \leq 1$.
3. **Arithmetic Mean Inequality:** $\dfrac{b_{yx} + b_{xy}}{2} \geq r$ (by AM $\geq$ GM).
4. **Centroid Intersection:** Both lines intersect at the point of means $(\bar{x}, \bar{y})$.
5. **Scale Dependence:** Regression coefficients are invariant to change of origin, but **depend** on scale: $b_{yx} = \frac{k}{h}b_{vu}$.

3.3 Angle Between the Two Regression Lines

Derivation: Angle θ Between Regression Lines

The slopes of the two lines in the (X,Y) plane are:

$$m_1 = b_{yx} = r\frac{\sigma_y}{\sigma_x}, \qquad m_2 = \frac{1}{b_{xy}} = \frac{\sigma_y}{r\sigma_x} \tag{182}$$

The angle θ between the lines satisfies:

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{\frac{\sigma_y}{r\sigma_x} - r\frac{\sigma_y}{\sigma_x}}{1 + \left(r\frac{\sigma_y}{\sigma_x}\right)\left(\frac{\sigma_y}{r\sigma_x}\right)}\right| = \left|\frac{\frac{\sigma_y}{\sigma_x}\left(\frac{1-r^2}{r}\right)}{1 + \frac{\sigma_y^2}{\sigma_x^2}}\right| \tag{183}$$

Simplifying the denominator $1 + \sigma_y^2/\sigma_x^2 = (\sigma_x^2 + \sigma_y^2)/\sigma_x^2$:

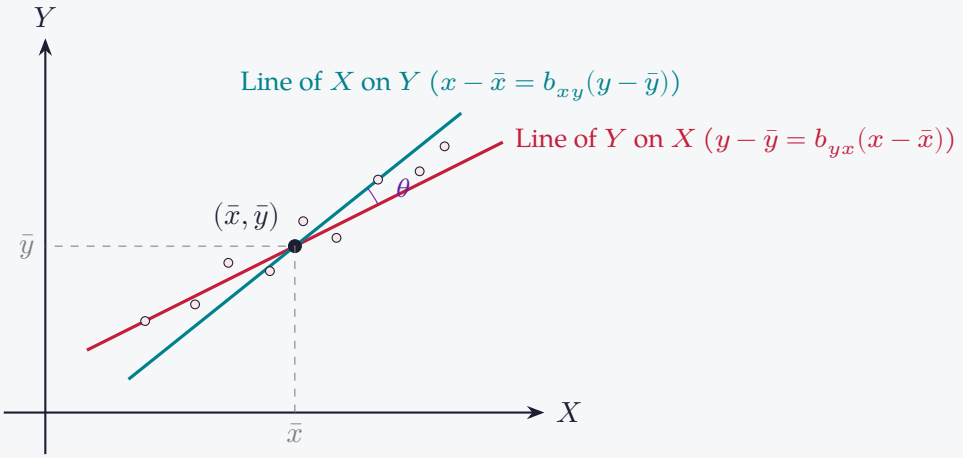
$$\tan\theta = \left(\frac{1 - r^2}{|r|}\right)\left(\frac{\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2}\right) \tag{184}$$

Special Cases:

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- If $r = \pm 1 \implies \tan\theta = 0 \implies \theta = 0^\circ$ (the two lines **coincide**; perfect linear correlation).
- If $r = 0 \implies \tan\theta \to \infty \implies \theta = 90^\circ$ (the two lines are **perpendicular** parallel to coordinate axes; uncorrelated).

TikZ Plot: Scatter Diagram and the Two Lines of Regression with Intersection at Centroid



3.4 Spearman's Rank Correlation Coefficient (r_s)

When variables represent qualitative attributes that cannot be measured numerically but can be ordered or ranked (e.g., beauty, intelligence, performance rankings):

Spearman's Rank Correlation Formula

Let R_{xi} and R_{yi} denote the ranks assigned to the i-th individual, with difference $d_i = R_{xi} - R_{yi}$.

1. **Case 1: When Ranks are Distinct (No Ties):**

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)} \tag{185}$$

2. **Case 2: When Ranks are Tied (Correction Factor):** If an attribute value repeats m times, all tied entries are allocated the average rank, and a correction factor $\frac{m(m^2-1)}{12}$ is added for each tie group:

$$r_s = 1 - \frac{6\left[\sum_{i=1}^{n} d_i^2 + \sum_{j} \frac{m_j(m_j^2-1)}{12}\right]}{n(n^2-1)} \tag{186}$$

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Quick Revision — Module 4

Basic Statistics, Moments, Distributions, Correlation and Regression | BS-M301

Key Formulas and Statistical Identities at a Glance

Empirical Mean-Mode	$\text{Mode} \approx 3\text{Median} - 2\text{Mean} \iff \text{Mean} - \text{Mode} \approx 3(\text{Mean} - \text{Median})$
Central from Raw Moments	$\mu_2 = \mu_2' - (\mu_1')^2,\ \mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3,\ \mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$
Pearson Skewness	$S_k = \frac{\text{Mean}-\text{Mode}}{\sigma} = \frac{3(\text{Mean}-\text{Median})}{\sigma},\quad \beta_1 = \frac{\mu_3^2}{\mu_2^3},\quad \gamma_1 = \frac{\mu_3}{\sigma^3}$
Kurtosis Coefficients	$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4},\quad \gamma_2 = \beta_2 - 3\quad (\text{Meso} = 3, \text{Lepto} > 3, \text{Platy} < 3)$
Binomial Parameters	$\mu = np,\quad \sigma^2 = npq,\quad \beta_1 = \frac{(q-p)^2}{npq},\quad \beta_2 = 3 + \frac{1-6pq}{npq}$
Poisson Parameters	$\mu = \lambda,\quad \sigma^2 = \lambda,\quad \beta_1 = \frac{1}{\lambda},\quad \beta_2 = 3 + \frac{1}{\lambda}$
Normal Moments	$\text{All odd } \mu_{2k+1} = 0,\quad \mu_{2k} = \frac{(2k)!}{2^k k!}\sigma^{2k},\quad \beta_1 = 0,\quad \beta_2 = 3$
Pearson Correlation	$r = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}} \in [-1, +1]$
Regression Lines	$Y - \bar{y} = b_{yx}(X - \bar{x}),\quad X - \bar{x} = b_{xy}(Y - \bar{y}),\quad b_{yx} = r\frac{\sigma_y}{\sigma_x},\ b_{xy} = r\frac{\sigma_x}{\sigma_y}$
Regression Properties	$r = \pm\sqrt{b_{yx} b_{xy}},\quad \frac{b_{yx}+b_{xy}}{2} \geq r,\quad \tan\theta = \frac{1-r^2}{\lvert r\rvert}\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$
Spearman Rank Corr.	$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}\quad (\text{Ties add } \sum \frac{m_j(m_j^2-1)}{12})$

One-Line Definitions

- **Arithmetic Mean:** Center of gravity of data where algebraic sum of deviations $\sum(x_i - \bar{x})$ is identically zero.
- **Median & Mode:** Median is positional 50th percentile; Mode is observation exhibiting peak frequency density.
- **Skewness:** Metric quantifying directional asymmetry of frequency curve tail relative to central peak.
- **Kurtosis:** Measure of peakedness and tail fatness relative to standard Gaussian benchmark ($\beta_2 = 3$).
- **Leptokurtic Curve:** Sharply peaked frequency profile with fat heavy tails exhibiting $\beta_2 > 3$.
- **Platykurtic Curve:** Broad-shouldered flattened frequency curve with light rapidly decaying tails ($\beta_2 < 3$).
- **Pearson's r:** Standardized dimensionless measure of the strength and direction of linear association.
- **Regression Line:** Optimal line minimizing sum of squared vertical/horizontal prediction errors (Least Squares).
- **Regression Coefficients:** Directional slopes b_{yx} and b_{xy} whose geometric mean equals correlation coefficient r.
- **Spearman's Rank Correlation:** Non-parametric correlation measuring monotonic agreement between ordinal rankings.

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Frequently Asked / Viva Questions (Part 1)

Q1. State the algebraic properties of the Arithmetic Mean.

(1) The sum of deviations of all observations from their arithmetic mean is zero: $\sum f_i(x_i - \bar{x}) = 0$. (2) The sum of squared deviations is minimal when taken about the mean: $\sum f_i(x_i - A)^2$ is minimum for $A = \bar{x}$. (3) Mean of combined series: $\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1+n_2}$.

Q2. Derive the relationship $\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$.

$\mu_3 = E[(X-\mu)^3] = E[X^3 - 3X^2\mu + 3X\mu^2 - \mu^3] = E[X^3] - 3\mu E[X^2] + 3\mu^2 E[X] - \mu^3$. Substituting $E[X] = \mu'_1, E[X^2] = \mu'_2, E[X^3] = \mu'_3$: $\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 3(\mu'_1)^3 - (\mu'_1)^3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$.

Q3. Distinguish between positive skewness and negative skewness with sketches.

In positive skewness, the tail is extended to the right, and Mean $>$ Median $>$ Mode ($\gamma_1 > 0$). In negative skewness, the tail is extended to the left, and Mode $>$ Median $>$ Mean ($\gamma_1 < 0$).

Q4. What are Leptokurtic, Mesokurtic and Platykurtic curves?

A curve with $\beta_2 = 3$ is Mesokurtic (Normal). If $\beta_2 > 3$, it is Leptokurtic (more peaked with heavier tails). If $\beta_2 < 3$, it is Platykurtic (flatter top with lighter tails).

Q5. State the recurrence relation for central moments of a Poisson distribution.

$\mu_{r+1} = \lambda\left[r\mu_{r-1} + \frac{d\mu_r}{d\lambda}\right]$. For $r = 1$: $\mu_2 = \lambda(1 \cdot 0 + 1) = \lambda$. For $r = 2$: $\mu_3 = \lambda(2 \cdot 0 + \frac{d\lambda}{d\lambda}) = \lambda$. For $r = 3$: $\mu_4 = \lambda(3\lambda + \frac{d\lambda}{d\lambda}) = 3\lambda^2 + \lambda$.

Q6. Prove that the correlation coefficient is the geometric mean of the two regression coefficients.

$b_{yx} = r\frac{\sigma_y}{\sigma_x}$ and $b_{xy} = r\frac{\sigma_x}{\sigma_y}$. Multiplying: $b_{yx} \cdot b_{xy} = \left(r\frac{\sigma_y}{\sigma_x}\right)\left(r\frac{\sigma_x}{\sigma_y}\right) = r^2$. Taking the square root: $r = \pm\sqrt{b_{yx} \cdot b_{xy}}$.

Q7. Can two regression coefficients be 1.5 and 0.8? Explain.

Yes! $b_{yx} \cdot b_{xy} = (1.5)(0.8) = 1.2 > 1$. But wait: $r^2 = b_{yx}b_{xy} \leq 1$. Since $1.2 > 1$, this would imply $r^2 > 1$, which violates the Cauchy-Schwarz bound $|r| \leq 1$. Hence $b_{yx} = 1.5$ and $b_{xy} = 0.8$ is impossible!

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Frequently Asked / Viva Questions (Part 2)

Q8. Why are there two lines of regression rather than a single line?

Because the regression of Y on X minimizes the sum of squared vertical errors $\sum(y_i - \hat{y}_i)^2$, whereas the regression of X on Y minimizes the sum of squared horizontal errors $\sum(x_i - \hat{x}_i)^2$. The two lines coincide if and only if $r = \pm 1$ (zero error).

Q9. State the formula for the angle θ between the two regression lines and interpret the extreme cases.

$\tan\theta = \frac{1-r^2}{|r|}\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$. If $r = \pm 1$, $\tan\theta = 0 \implies \theta = 0^\circ$ (lines coincide; perfect correlation). If $r = 0$, $\tan\theta = \infty \implies \theta = 90^\circ$ (lines are perpendicular; uncorrelated).

Q10. Derive Spearman's Rank Correlation formula when ranks are not tied.

Let x_i, y_i be ranks from 1 to n. Then $\bar{x} = \bar{y} = \frac{n+1}{2}$, $\sigma_x^2 = \sigma_y^2 = \frac{n^2-1}{12}$. Since $d_i = x_i - y_i$, $\sum d_i^2 = \sum(x_i-\bar{x})^2 + \sum(y_i-\bar{y})^2 - 2\sum(x_i-\bar{x})(y_i-\bar{y}) = 2n\sigma_x^2 - 2nr\sigma_x^2 = 2n\sigma_x^2(1-r_s)$. Substituting $\sigma_x^2 = \frac{n^2-1}{12}$ yields $r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$.

Summary Matrix: Comparison of Binomial, Poisson and Normal Distributions

Parameter	Binomial $\text{Bin}(n,p)$	Poisson $\text{Poisson}(\lambda)$	Normal $\mathcal{N}(\mu,\sigma^2)$	Comparative Property
Mean ($E[X]$)	np	λ	μ	Location measure.
Variance (σ^2)	$npq < \mu$	$\lambda = \mu$	σ^2 (**Independent of** μ)	$\text{Var}_{\text{Bin}} < \text{Mean}$; $\text{Var}_{\text{Poi}} = \text{Mean}$.
MGF ($M_X(t)$)	$(q+pe^t)^n$	$\exp(\lambda(e^t - 1))$	$\exp(\mu t + \frac{1}{2}\sigma^2 t^2)$	Poisson is limiting Binomial MGF.
Skewness (γ_1)	$\frac{q-p}{\sqrt{npq}}$	$\frac{1}{\sqrt{\lambda}}$	0	Binomial is symmetric if $p = 0.5$.
Kurtosis (β_2)	$3 + \frac{1-6pq}{npq}$	$3 + \frac{1}{\lambda}$	3 ($\gamma_2 = 0$)	Both converge to 3 as $n, \lambda \to \infty$.

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Module 5

Curve Fitting by the Method of Least Squares

1. Curve Fitting by the Method of Least Squares

In experimental electronics and engineering measurement, observational data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ rarely lie on an exact mathematical curve due to sensor noise and environmental fluctuations. The **Method of Least Squares** provides an objective criterion to establish the best-fitting functional relationship $y = f(x; a, b, c, \dots)$ by minimizing the sum of the squared vertical discrepancies.

Definition: The Principle of Least Squares

Let (x_i, y_i) be n paired experimental measurements. For any chosen curve $y = f(x)$, the residual (vertical error) for the i-th observation is:

$$e_i = y_i - f(x_i) \tag{187}$$

The Principle of Least Squares states that the optimal curve parameters are those that minimize the total sum of squared residuals:

$$S = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - f(x_i; a, b, c, \dots)]^2 = \text{Minimum} \tag{188}$$

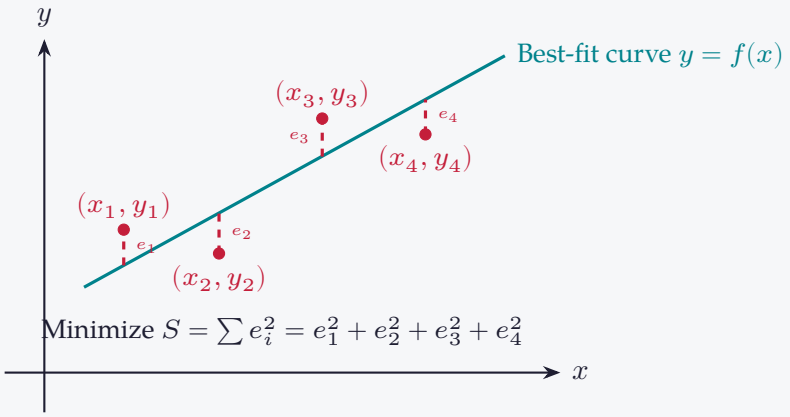
To achieve this minimum, the partial derivatives of S with respect to each undetermined parameter must vanish simultaneously:

$$\frac{\partial S}{\partial a} = 0, \qquad \frac{\partial S}{\partial b} = 0, \qquad \frac{\partial S}{\partial c} = 0, \quad \dots \tag{189}$$

The resulting system of simultaneous equations is called the **Normal Equations**.

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TikZ Schematic: Residual Errors and Best-Fit Curve in Least Squares



1.1 1. Fitting of a Straight Line: $y = a + bx$

Derivation of Normal Equations for a Straight Line

Let the straight line equation be $y = a + bx$. The error sum of squares is:

$$S = \sum_{i=1}^{n} (y_i - a - bx_i)^2 \tag{190}$$

Differentiating partially with respect to a and equating to zero:

$$\frac{\partial S}{\partial a} = \sum_{i=1}^{n} 2(y_i - a - bx_i)(-1) = 0 \implies \sum_{i=1}^{n} (y_i - a - bx_i) = 0$$

$$\implies \sum y = na + b \sum x \tag{191}$$

Differentiating partially with respect to b and equating to zero:

$$\frac{\partial S}{\partial b} = \sum_{i=1}^{n} 2(y_i - a - bx_i)(-x_i) = 0 \implies \sum_{i=1}^{n} x_i(y_i - a - bx_i) = 0$$

$$\implies \sum xy = a \sum x + b \sum x^2 \tag{192}$$

The Normal Equations for a Straight Line:

$$\sum y = na + b \sum x \tag{193}$$

$$\sum xy = a \sum x + b \sum x^2 \tag{194}$$

Computation Shortcut (Change of Origin): If n is odd and values of x are equally spaced, choosing the origin at the midpoint such that $\sum u = 0$ simplifies the solution to:

$$a = \frac{\sum y}{n}, \qquad b = \frac{\sum uy}{\sum u^2} \tag{195}$$

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1.2 2. Fitting of a Second-Degree Parabola: $y = a + bx + cx^2$

Normal Equations for a Second-Degree Parabola

For the quadratic curve $y = a + bx + cx^2$, the sum of squared errors is $S = \sum(y_i - a - bx_i - cx_i^2)^2$. Setting $\frac{\partial S}{\partial a} = 0$, $\frac{\partial S}{\partial b} = 0$, $\frac{\partial S}{\partial c} = 0$ yields the 3×3 system of **Normal Equations**:

$$\sum y = na + b\sum x + c\sum x^2 \tag{196}$$

$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3 \tag{197}$$

$$\sum x^2 y = a\sum x^2 + b\sum x^3 + c\sum x^4 \tag{198}$$

When origin is shifted to make $\sum u = 0$ (which also implies $\sum u^3 = 0$ for symmetric spacing):

$$\sum y = na + c\sum u^2 \tag{199}$$

$$\sum uy = b\sum u^2 \implies b = \frac{\sum uy}{\sum u^2} \tag{200}$$

$$\sum u^2 y = a\sum u^2 + c\sum u^4 \tag{201}$$

1.3 3. Fitting of Non-Linear / General Curves via Linearization

Many non-linear physical relationships can be linearized by taking logarithms:

Curve Model	Log Transformation	Transformed Form	Normal Equations
Exponential Curve $y = ae^{bx}$	$\ln y = \ln a + bx$	$Y = A + bx$ ($Y = \ln y, A = \ln a$)	$\sum Y = nA + b\sum x$ $\sum xY = A\sum x + b\sum x^2$
Power / Geometric $y = ax^b$	$\log_{10} y = \log_{10} a + b\log_{10} x$	$Y = A + bX$ ($Y = \log y, X = \log x$)	$\sum Y = nA + b\sum X$ $\sum XY = A\sum X + b\sum X^2$
Modified Exponential $y = ab^x$	$\log y = \log a + x\log b$	$Y = A + Bx$ ($A = \log a, B = \log b$)	$\sum Y = nA + B\sum x$ $\sum xY = A\sum x + B\sum x^2$
Hyperbolic Curve $y = \frac{a}{x} + b$	**Substitute** $u = \frac{1}{x}$	$y = au + b$	$\sum y = a\sum u + nb$ $\sum uy = a\sum u^2 + b\sum u$

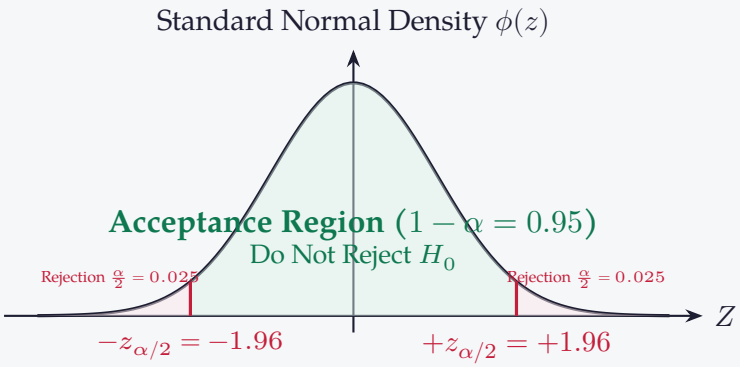
2. Foundations of Statistical Hypothesis Testing

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Definition: Terminology of Statistical Testing

1. **Statistical Hypothesis:** An assertion or conjecture concerning the unknown parameter(s) of a population distribution.
2. **Null Hypothesis (H_0):** The baseline hypothesis of *no effect*, *no difference*, or *status quo*. It is maintained until evidence indicates otherwise.
3. **Alternative Hypothesis (H_1):** The mutually exclusive claim that contradicts H_0. It can be **two-tailed** ($H_1 : \theta \neq \theta_0$) or **one-tailed** ($H_1 : \theta > \theta_0$ or $H_1 : \theta < \theta_0$).
4. **Errors in Hypothesis Testing:**
   - **Type I Error (α):** Rejecting H_0 when H_0 is actually true (Producer's risk / False positive).
   $$P(\text{Type I Error}) = \alpha = \text{Level of Significance} \tag{202}$$
   - **Type II Error (β):** Failing to reject H_0 when H_0 is actually false (Consumer's risk / False negative).
   - **Power of the Test ($1-\beta$):** Probability of correctly rejecting a false null hypothesis.
5. **Critical Region (Rejection Region):** The set of values of the test statistic for which H_0 is rejected.

TikZ Schematic: Standard Normal Test Statistic with Two-Tailed Critical Regions ($\alpha = 0.05$)



2.1 Critical Values for Large Sample Tests (Z-Distribution)

Level of Significance (α)	Two-Tailed Critical Value ($\lvert Z\rvert > z_{\alpha/2}$)	Right-Tailed Critical Value ($Z > z_{\alpha}$)	Left-Tailed Critical Value ($Z < -z_{\alpha}$)
$\alpha = 5\%$ (0.05)	$z_{0.025} = 1.96$	$z_{0.05} = 1.645$	$-z_{0.05} = -1.645$
$\alpha = 1\%$ (0.01)	$z_{0.005} = 2.576$	$z_{0.01} = 2.326$	$-z_{0.01} = -2.326$
$\alpha = 10\%$ (0.10)	$z_{0.05} = 1.645$	$z_{0.10} = 1.282$	$-z_{0.10} = -1.282$

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3. Large Sample Tests of Significance (Z-Tests, $n \geq 30$)

By the Central Limit Theorem, for sample sizes $n \geq 30$, sample statistics (proportions, means, variances) are approximately normally distributed regardless of the underlying parent population distribution. The general test statistic is:

$$Z = \frac{\text{Sample Statistic} - \text{Hypothesized Parameter Value}}{\text{Standard Error of the Statistic (SE)}} \sim \mathcal{N}(0,1) \tag{203}$$

3.1 1. Large Sample Test for a Single Proportion

Testing whether a population proportion P equals a specified value P_0.

Test for Single Proportion

- **Hypotheses:** $H_0 : P = P_0 \quad \text{vs} \quad H_1 : P \neq P_0$ (two-tailed) or $P > P_0$ (right-tailed).
- **Sample Proportion:** $p = \frac{x}{n}$, where x is the number of successes in n trials.
- **Standard Error under H_0:** $\text{SE}(p) = \sqrt{\frac{P_0 Q_0}{n}}$, where $Q_0 = 1 - P_0$.
- **Test Statistic:**

$$Z = \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \tag{204}$$

- **Decision Rule:** If $|Z| > 1.96$ (at 5% level), reject H_0.

3.2 2. Large Sample Test for Difference of Two Proportions

Testing whether two independent populations have identical success proportions ($P_1 = P_2$).

Test for Difference of Two Proportions

- **Hypotheses:** $H_0 : P_1 = P_2 \quad \text{vs} \quad H_1 : P_1 \neq P_2$.
- **Sample Proportions:** $p_1 = \frac{x_1}{n_1}$ and $p_2 = \frac{x_2}{n_2}$.
- **Pooled Proportion Estimate (under H_0):**

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \qquad \hat{q} = 1 - \hat{p} \tag{205}$$

- **Test Statistic:**

$$Z = \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \tag{206}$$

3.3 3. Large Sample Test for a Single Mean

Testing whether a population mean μ equals a specified value μ_0.

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Test for Single Mean

- **Hypotheses:** $H_0 : \mu = \mu_0 \quad \text{vs} \quad H_1 : \mu \neq \mu_0$.
- **Sample Mean:** $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$.
- **Test Statistic:**

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \tag{207}$$

If the population standard deviation σ is unknown, it is replaced by sample SD $s = \sqrt{\frac{1}{n}\sum(x_i - \bar{x})^2}$ (valid since $n \geq 30$):

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \tag{208}$$

3.4 4. Large Sample Test for Difference of Two Means

Testing whether two independent populations have equal means ($\mu_1 = \mu_2$).

Test for Difference of Two Means

- **Hypotheses:** $H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_1 \neq \mu_2$.
- **Test Statistic:**

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \approx \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{209}$$

3.5 5. Large Sample Test for Difference of Standard Deviations

Testing whether two populations have identical variability ($\sigma_1 = \sigma_2$).

Test for Difference of Two Standard Deviations

For a sample of size n drawn from a normal population, the sampling distribution of the sample standard deviation s is asymptotically normal with mean σ and standard error:

$$\text{SE}(s) = \frac{\sigma}{\sqrt{2n}} \tag{210}$$

Test Statistic for $H_0 : \sigma_1 = \sigma_2$:

$$Z = \frac{s_1 - s_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}} \approx \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} \tag{211}$$

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Quick Revision — Module 5

Applied Statistics, Curve Fitting and Tests of Significance | BS-M301

Key Normal Equations and Large Sample Z-Test Formulas at a Glance

Line Normal Equations	$\sum y = na + b\sum x, \quad \sum xy = a\sum x + b\sum x^2$
Parabola Normal Eqs.	$\sum y = na + b\sum x + c\sum x^2, \ \sum xy = a\sum x + b\sum x^2 + c\sum x^3, \ \sum x^2 y = a\sum x^2 + b\sum x^3 + c\sum x^4$
Exponential Curve	$y = ae^{bx} \implies \ln y = \ln a + bx \implies Y = A + bx$
Power Curve	$y = ax^b \implies \log y = \log a + b\log x \implies Y = A + bX$
Single Proportion Test	$Z = \frac{p - P_0}{\sqrt{P_0 Q_0/n}}, \quad Q_0 = 1 - P_0$
Two Proportions Test	$Z = \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}(1/n_1 + 1/n_2)}}, \quad \hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \ \hat{q} = 1 - \hat{p}$
Single Mean Test	$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \approx \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
Two Means Test	$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \approx \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$
Standard Deviation SE	$\mathrm{SE}(s) = \frac{\sigma}{\sqrt{2n}}, \quad Z = \frac{s_1 - s_2}{\sqrt{s_1^2/(2n_1) + s_2^2/(2n_2)}}$
Critical Z-Values	$\alpha = 5\% : \lvert Z\rvert = 1.96$ (two-tailed), 1.645 (one-tailed); $\alpha = 1\% : \lvert Z\rvert = 2.576$

One-Line Definitions

- **Least Squares Principle:** Criterion minimizing the sum of squared vertical residuals $\sum(y_i - f(x_i))^2$.
- **Normal Equations:** Set of simultaneous linear equations obtained by setting partial derivatives of error to zero.
- **Null Hypothesis (H_0):** Initial baseline claim stating no difference, no effect, or parameter equality.
- **Alternative Hypothesis (H_1):** The competing research claim adopted if sample evidence contradicts H_0.
- **Type I Error (α):** Mistakenly rejecting a true null hypothesis (false positive; significance level).
- **Type II Error (β):** Incorrectly failing to reject a false null hypothesis (false negative).
- **Power of Test ($1 - \beta$):** Probability of correctly rejecting H_0 when the alternative H_1 is true.
- **Critical Region:** Subspace of test statistics leading to rejection of H_0 at significance level α.
- **Standard Error (SE):** Standard deviation of the sampling distribution of a given sample statistic.
- **Pooled Proportion ($\hat{p}$):** Combined success rate $(x_1 + x_2)/(n_1 + n_2)$ under the null hypothesis of equality.

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Frequently Asked / Viva Questions (Part 1)

Q1. State the Principle of Least Squares and explain why vertical errors are squared.

The principle minimizes $S = \sum e_i^2 = \sum (y_i - \hat{y}_i)^2$. Errors are squared because (1) positive and negative errors would otherwise cancel out ($\sum e_i = 0$), (2) squaring penalizes large outliers heavily, and (3) S is continuously differentiable, yielding linear normal equations.

Q2. How can the exponential curve $y = ab^x$ be fitted using the method of least squares?

Taking base-10 logarithms: $\log y = \log a + x \log b$. Defining $Y = \log y$, $A = \log a$, and $B = \log b$, the model becomes the linear form $Y = A + Bx$. Solving standard linear normal equations for A and B, the parameters are recovered via $a = 10^A$ and $b = 10^B$.

Q3. Distinguish between Type I Error and Type II Error in hypothesis testing.

Type I error (α) occurs when a true null hypothesis H_0 is mistakenly rejected (false alarm). Type II error (β) occurs when a false null hypothesis is accepted (missed detection). Decreasing α for a fixed sample size increases β.

Q4. Why can the normal distribution be used for large sample tests even if the population is non-normal?

By the Central Limit Theorem (CLT), the sampling distribution of the sample mean $\bar{x}$ (and sample proportions) approaches a normal distribution as sample size $n \geq 30$, irrespective of the shape of the parent population.

Q5. What is the difference between a one-tailed test and a two-tailed test?

A two-tailed test investigates departures in either direction ($H_1 : \mu \neq \mu_0$), placing critical area $\alpha/2$ in both tails. A one-tailed test investigates directional superiority or inferiority ($H_1 : \mu > \mu_0$ or $\mu < \mu_0$), concentrating the entire critical area α in one tail.

Q6. State the standard error of the sample standard deviation for a normal population.

For large samples drawn from a normal population, $\text{SE}(s) = \frac{\sigma}{\sqrt{2n}}$. It is used to test significance of variance differences across industrial production lines.

Q7. Why do we pool the proportions $\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ in the two-proportion test?

Under the null hypothesis $H_0 : P_1 = P_2 = P$, both samples originate from a single population with common proportion P. Pooling gives the minimum-variance unbiased estimator of this common parameter.

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Frequently Asked / Viva Questions (Part 2)

Q8. What decision is made if the calculated test statistic is $Z_{\text{cal}} = 2.15$ at 5% significance level?

For a two-tailed test at $\alpha = 5\%$, the critical boundary is $Z_{\text{crit}} = 1.96$. Since $|Z_{\text{cal}}| = 2.15 > 1.96$, the sample statistic falls in the critical rejection region. We reject the null hypothesis H_0 and conclude the observed difference is statistically significant.

Q9. How does shifting the origin simplify fitting a parabola $y = a + bx + cx^2$?

If x values are symmetrically spaced, shifting the origin to the center makes $\sum u = 0$ and $\sum u^3 = 0$. The normal equations decouple into $b = \sum uy / \sum u^2$, while a and c decouple from b into an elementary 2×2 system, dramatically reducing computational effort.

Q10. What is the Level of Significance (α) and how is it related to the confidence level?

The level of significance α is the maximum tolerable probability of committing a Type I error. It is related to the confidence level C by $C = (1 - \alpha) \times 100\%$. For example, an $\alpha = 0.05$ level of significance corresponds to a 95% confidence level.

Summary Matrix: Comprehensive Decision Table of Large Sample Significance Tests

Test Type	Null Hypothesis (H_0)	Standard Error (SE)	Test Statistic (Z_{cal})
Single Proportion	$H_0 : P = P_0$	$\sqrt{\frac{P_0 Q_0}{n}}$	$Z = \frac{p - P_0}{\sqrt{P_0 Q_0 / n}}$
Two Proportions	$H_0 : P_1 = P_2$	$\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$	$Z = \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}(1/n_1 + 1/n_2)}}$, $\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$
Single Mean	$H_0 : \mu = \mu_0$	$\frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$	$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
Two Means	$H_0 : \mu_1 = \mu_2$	$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$
Two Std. Deviations	$H_0 : \sigma_1 = \sigma_2$	$\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}$	$Z = \frac{s_1 - s_2}{\sqrt{s_1^2/(2n_1) + s_2^2/(2n_2)}}$

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Module 6
Introduction to Small Sample Theory

1. Introduction to Small Sample Theory

Definition: Small Samples vs. Large Samples

In statistical hypothesis testing, a sample is designated as a **small sample** if its sample size $n < 30$.

- **Large Samples ($n \geq 30$):** By the Central Limit Theorem (CLT), the sampling distribution of sample means approaches normality regardless of the parent distribution. The sample standard deviation s closely approximates the population standard deviation σ, justifying the use of the standard normal Z-test.
- **Small Samples ($n < 30$):** The CLT cannot be invoked. The parent population **must be assumed normally distributed**. Furthermore, replacing σ with the sample variance $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ introduces substantial additional sampling fluctuations, yielding distributions with heavier tails than $N(0,1)$.

Hence, small sample inference relies on the **exact sampling distributions**:

1. **Student's t-distribution** (W. S. Gosset, 1908): For inference on population means and correlation coefficients when σ is unknown.
2. **Snedecor's F-distribution** (G. W. Snedecor & R. A. Fisher): For testing the ratio of two independent population variances.
3. **Chi-square (χ^2) distribution** (Karl Pearson, 1900): For tests of goodness of fit and independence of attributes in contingency tables.

Degrees of Freedom (ν or d.f.)

The **degrees of freedom** (ν) of a test statistic represents the number of unconstrained, freely variable observations available in the calculation:

$$\nu = n - k \tag{212}$$

where n is the total number of sample observations and k is the number of linear constraints or sample-derived parameters estimated to compute the statistic (e.g., estimating μ by $\bar{x}$ consumes 1 d.f., so $\nu = n - 1$).

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2. Student's t-Distribution

2.1 Definition and Derivation

Mathematical Definition of Student's t-Variable

Let $Z \sim N(0,1)$ be a standard normal random variable, and let $Y \sim \chi^2(\nu)$ be an independent Chi-square random variable with ν degrees of freedom. The ratio:

$$T = \frac{Z}{\sqrt{\frac{Y}{\nu}}} \tag{213}$$

follows **Student's t-distribution** with ν degrees of freedom, written $T \sim t(\nu)$.

2.2 Probability Density Function and Properties

The probability density function (PDF) of $T \sim t(\nu)$ is:

$$f(t) = \frac{1}{\sqrt{\nu}\,\mathrm{B}\left(\frac{1}{2}, \frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad -\infty < t < \infty \tag{214}$$

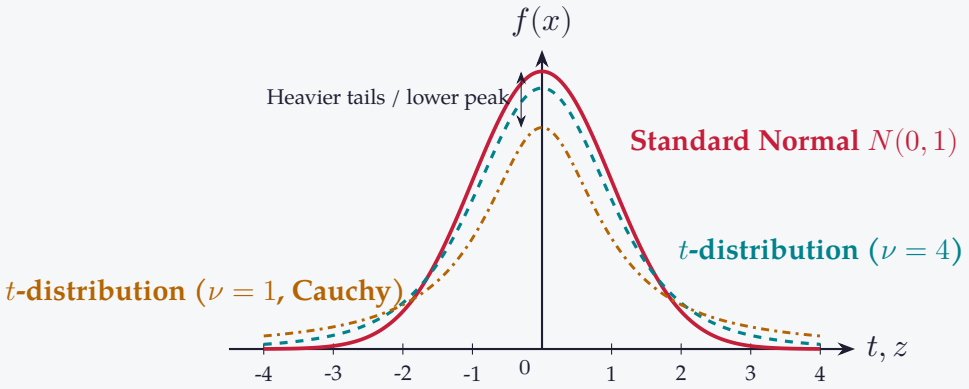
where $\mathrm{B}\left(\frac{1}{2}, \frac{\nu}{2}\right) = \frac{\sqrt{\pi}\,\Gamma(\nu/2)}{\Gamma((\nu+1)/2)}$ is the Beta function.

Key Properties of Student's t-Distribution

1. **Symmetry:** The curve is bell-shaped and symmetric about $t = 0$. All odd central moments are zero:
$$\mu_{2k+1} = E[T^{2k+1}] = 0 \quad (\text{for } \nu > 2k+1) \tag{215}$$
2. **Mean:** $\mu_1' = E[T] = 0$ (valid for $\nu > 1$).
3. **Variance:** $\mathrm{Var}(T) = \mu_2 = \dfrac{\nu}{\nu - 2}$ (valid for $\nu > 2$). Note that $\mathrm{Var}(T) > 1$.
4. **Kurtosis (Heavy Tails):** The distribution is leptokurtic ($\beta_2 > 3$). For $\nu > 4$:
$$\beta_2 = \frac{\mu_4}{\mu_2^2} = 3 + \frac{6}{\nu - 4} > 3 \tag{216}$$
5. **Limiting Form (Asymptotic Normality):** As $\nu \to \infty$, the t-distribution converges to the standard normal distribution:
$$\lim_{\nu\to\infty} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}} = e^{-t^2/2} \implies t(\nu) \xrightarrow{d} N(0,1) \tag{217}$$

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TikZ Illustration: Comparison of Student's t vs. Standard Normal $N(0,1)$



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3. Student's t-Tests of Significance

3.1 Test for Single Mean

Test 1: Single Mean t-Test Formulation

Let $x_1, x_2, \ldots, x_n$ be a random sample of size $n < 30$ drawn from a normal population $N(\mu, \sigma^2)$ where σ is unknown. We test:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0 \quad (\text{or one-sided } \mu > \mu_0, \mu < \mu_0) \tag{218}$$

The test statistic is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \tag{219}$$

where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and the **unbiased sample variance** is:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - \frac{(\sum x_i)^2}{n}\right) \tag{220}$$

Under H_0, $t \sim t(\nu)$ with degrees of freedom $\nu = n - 1$.
Decision Rule (Two-tailed at significance level α):

- Reject H_0 if $|t_{\text{cal}}| > t_{\alpha/2, n-1}$.
- Fail to reject H_0 if $|t_{\text{cal}}| \leq t_{\alpha/2, n-1}$.

$100(1-\alpha)\%$ **Confidence Interval for μ:**

$$\bar{x} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \tag{221}$$

Worked Example: Single Mean t-Test

A manufacturer claims that the mean lifetime of an electronic component is 1000 hours. A random sample of 9 components yields a sample mean of 980 hours with a sample standard deviation $s = 30$ hours. Test at the 5% level whether the true mean lifetime is significantly less than 1000 hours ($t_{0.05,8} = 1.860$).
Solution:

1. **Hypotheses:** $H_0 : \mu = 1000$ vs. $H_1 : \mu < 1000$ (one-tailed left-sided test).
2. **Parameters:** $n = 9$, $\bar{x} = 980$, $s = 30$, $\mu_0 = 1000$.
3. **Test Statistic:**
$$t_{\text{cal}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{980 - 1000}{30/\sqrt{9}} = \frac{-20}{10} = -2.00$$
4. **Critical Value:** For left-tailed test at $\alpha = 0.05$ with $\nu = 8$: $t_{\text{crit}} = -1.860$.
5. **Conclusion:** Since $t_{\text{cal}} = -2.00 < -1.860$, t_{cal} falls in the critical rejection region. We reject H_0 and conclude that the mean lifetime is significantly less than 1000 hours.

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3.2 Test for Difference of Two Means (Independent Samples)

Test 2: Two-Sample Independent t-Test with Pooled Variance

Let two independent samples of sizes n_1 and n_2 be drawn from normal populations $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$ with **equal unknown variance** ($\sigma_1^2 = \sigma_2^2 = \sigma^2$). We test:

$$H_0 : \mu_1 = \mu_2 \quad \text{vs.} \quad H_1 : \mu_1 \neq \mu_2 \tag{222}$$

The **pooled unbiased variance** s_p^2 combining both samples is:

$$s_p^2 = \frac{\sum_{i=1}^{n_1}(x_{1i} - \bar{x}_1)^2 + \sum_{j=1}^{n_2}(x_{2j} - \bar{x}_2)^2}{n_1 + n_2 - 2} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \tag{223}$$

The test statistic is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}} \tag{224}$$

Under H_0, $t \sim t(\nu)$ with degrees of freedom $\nu = n_1 + n_2 - 2$.

3.3 Paired t-Test for Dependent Samples

Test 3: Paired t-Test Principle

When observations in two samples are drawn in matched pairs (e.g., measurements taken on the **same physical circuit before and after** temperature cycling), the two groups are statistically dependent. We eliminate inter-subject variability by evaluating individual differences:

$$d_i = x_i - y_i, \quad i = 1, 2, \ldots, n \tag{225}$$

Under the null hypothesis of no treatment effect ($H_0 : \mu_d = 0$):

$$t = \frac{\bar{d} - 0}{s_d/\sqrt{n}} \tag{226}$$

where:

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \qquad s_d^2 = \frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2 = \frac{\sum d_i^2 - \dfrac{(\sum d_i)^2}{n}}{n-1} \tag{227}$$

The statistic follows Student's t-distribution with $\nu = n - 1$ degrees of freedom.

3.4 Test for Significance of Correlation Coefficient

Test 4: Significance of Correlation Coefficient

Let r be the Pearson sample correlation coefficient calculated from n pairs of observations drawn from a bivariate normal population. To test:

$$H_0 : \rho = 0 \quad \text{(variables are uncorrelated)} \quad \text{vs.} \quad H_1 : \rho \neq 0 \tag{228}$$

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The test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \tag{229}$$

Under H_0, $t \sim t(\nu)$ with degrees of freedom $\nu = n - 2$. Reject H_0 if $|t_{\text{cal}}| > t_{\alpha/2, n-2}$.

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4. Snedecor's F-Distribution and Test for Ratio of Variances

4.1 Definition and Properties

Definition of Snedecor's F-Distribution

Let $U \sim \chi^2(\nu_1)$ and $V \sim \chi^2(\nu_2)$ be two independent Chi-square random variables with ν_1 and ν_2 degrees of freedom, respectively. The ratio:

$$F = \frac{U/\nu_1}{V/\nu_2} \tag{230}$$

follows **Snedecor's F-distribution** with (ν_1, ν_2) degrees of freedom, denoted $F \sim F(\nu_1, \nu_2)$.

Properties of the F-Distribution

1. **Support and Shape:** $F \in (0, \infty)$. The distribution is positively skewed (right-skewed) and unimodal.
2. **Mean:** $E[F] = \dfrac{\nu_2}{\nu_2 - 2}$ (for $\nu_2 > 2$, independent of ν_1).
3. **Variance:** $\mathrm{Var}(F) = \dfrac{2\nu_2^2(\nu_1 + \nu_2 - 2)}{\nu_1(\nu_2 - 2)^2(\nu_2 - 4)}$ (for $\nu_2 > 4$).
4. **Reciprocal Property:** If $F \sim F(\nu_1, \nu_2)$, then $\dfrac{1}{F} \sim F(\nu_2, \nu_1)$. Hence:

$$F_{1-\alpha,\nu_1,\nu_2} = \frac{1}{F_{\alpha,\nu_2,\nu_1}} \tag{231}$$

5. **Relation to t and χ^2:**

$$F_{1,\nu} = t_\nu^2, \qquad \lim_{\nu_2 \to \infty} \nu_1 F_{\nu_1,\nu_2} = \chi^2(\nu_1) \tag{232}$$

4.2 F-Test for Equality of Two Population Variances

F-Test Procedure for $\sigma_1^2 = \sigma_2^2$

Let two independent random samples of sizes n_1 and n_2 be drawn from normal populations $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$. We test:

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad \text{vs.} \quad H_1 : \sigma_1^2 \neq \sigma_2^2 \quad (\text{or } \sigma_1^2 > \sigma_2^2) \tag{233}$$

Compute the unbiased sample variances:

$$s_1^2 = \frac{1}{n_1 - 1}\sum_{i=1}^{n_1}(x_{1i} - \bar{x}_1)^2, \qquad s_2^2 = \frac{1}{n_2 - 1}\sum_{j=1}^{n_2}(x_{2j} - \bar{x}_2)^2 \tag{234}$$

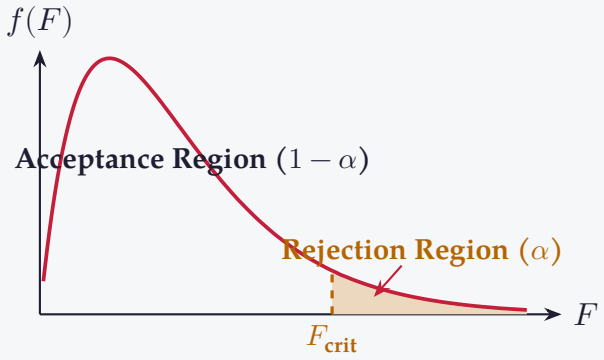
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By convention, we place the larger sample variance in the numerator:

$$F_{\text{cal}} = \frac{s_1^2}{s_2^2} \geq 1 \quad (\text{assuming } s_1^2 \geq s_2^2) \tag{235}$$

with degrees of freedom $\nu_1 = n_1 - 1$ (numerator) and $\nu_2 = n_2 - 1$ (denominator). Reject H_0 at level α if $F_{\text{cal}} > F_{\alpha/2,\nu_1,\nu_2}$.

TikZ Schematic: Right-Skewed F-Distribution with Critical Region



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5. Chi-Square (χ^2) Distribution

5.1 Definition and Properties

Definition of Chi-Square Distribution

If $Z_1, Z_2, \ldots, Z_\nu$ are ν independent standard normal random variables ($Z_i \sim N(0,1)$), the sum of their squares:

$$X = \sum_{i=1}^{\nu} Z_i^2 \tag{236}$$

follows a **Chi-square distribution** with ν degrees of freedom, denoted $X \sim \chi^2(\nu)$.

Mathematical Properties of $\chi^2(\nu)$

1. **Probability Density Function:**

$$f(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\frac{\nu}{2}-1} e^{-x/2}, \quad x \geq 0 \tag{237}$$

2. **Mean and Variance:**

$$E[X] = \nu, \qquad \text{Var}(X) = 2\nu \tag{238}$$

3. **Moment Generating Function (MGF):**

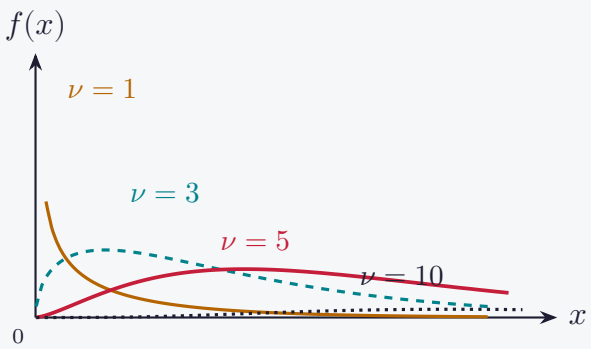
$$M_X(t) = (1-2t)^{-\nu/2}, \quad \text{for } t < \frac{1}{2} \tag{239}$$

4. **Additive (Reproductive) Property:** If $X_1 \sim \chi^2(\nu_1)$ and $X_2 \sim \chi^2(\nu_2)$ are independent:

$$X_1 + X_2 \sim \chi^2(\nu_1 + \nu_2) \tag{240}$$

5. **Limiting Form:** For large ν ($\nu > 30$), $\sqrt{2\chi^2} - \sqrt{2\nu - 1} \approx N(0,1)$.

TikZ Plot: Chi-Square Curves for Degrees of Freedom $\nu = 1, 3, 5, 10$



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6. Chi-Square Test of Goodness of Fit

6.1 Principle and Pearson's Test Statistic

Pearson's Chi-Square Goodness of Fit

The goodness of fit test evaluates whether an observed frequency distribution conforms to a theoretical probability distribution (such as Binomial, Poisson, or Normal). Let:

- O_i = Observed frequency in the i-th class ($i = 1, 2, \ldots, k$).
- E_i = Expected frequency under theoretical model, with $\sum_{i=1}^{k} E_i = \sum_{i=1}^{k} O_i = N$.

Karl Pearson proved that the test statistic:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = \sum_{i=1}^{k} \frac{O_i^2}{E_i} - N \tag{241}$$

asymptotically follows a Chi-square distribution with ν degrees of freedom:

$$\nu = k - 1 - m \tag{242}$$

where k is the number of classes after pooling, and m is the number of independent population parameters estimated from the sample data.

Conditions for Validity of χ^2 Test

To ensure accurate approximation of discrete multinomial probabilities by continuous χ^2:

1. **Total Sample Size:** $N = \sum O_i \geq 50$.
2. **Minimum Expected Frequency:** Each $E_i \geq 5$. If any $E_i < 5$, it must be **pooled (combined)** with adjacent classes, reducing k.
3. **Independent Observations:** Sampling units must be mutually independent.
4. **Linear Constraint:** $\sum O_i = \sum E_i = N$, which consumes 1 degree of freedom.

Worked Example: Chi-Square Test of Goodness of Fit

A die is rolled 120 times, yielding the following face frequencies: 15, 23, 18, 16, 26, 22. Test at the 5% level whether the die is fair ($\chi^2_{0.05,5} = 11.07$).
Solution:

1. **Hypotheses:** H_0: The die is fair ($p_i = 1/6$) vs. H_1: The die is biased.
2. **Expected Frequencies:** $E_i = 120 \times \frac{1}{6} = 20$ for all $i = 1, \ldots, 6$. All $E_i \geq 5$.
3. **Computation Table:**

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Face (i)	Observed (O_i)	Expected (E_i)	$O_i - E_i$	$(O_i - E_i)^2/E_i$
1	15	20	-5	$25/20 = 1.25$
2	23	20	$+3$	$9/20 = 0.45$
3	18	20	-2	$4/20 = 0.20$
4	16	20	-4	$16/20 = 0.80$
5	26	20	$+6$	$36/20 = 1.80$
6	22	20	$+2$	$4/20 = 0.20$
Total	**120**	**120**	**0**	$\chi^2_{\text{cal}} = \mathbf{4.70}$

Decision: With $\nu = 6 - 1 = 5$ d.f., $\chi^2_{\text{crit}} = 11.07$. Since $\chi^2_{\text{cal}} = 4.70 < 11.07$, we fail to reject H_0. The die is consistent with being unbiased.

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7. Chi-Square Test for Independence of Attributes

7.1 $r \times c$ Contingency Tables

Contingency Table and Expected Cell Frequency

When N observations are classified simultaneously across two qualitative attributes A (with r levels $A_1, \dots, A_r$) and B (with c levels $B_1, \dots, B_c$), the frequencies form an $r \times c$ contingency table.
Null Hypothesis: H_0: Attributes A and B are independent.
Under H_0, $P(A_i \cap B_j) = P(A_i)P(B_j)$, yielding the **expected cell frequency** E_{ij}:

$$E_{ij} = N \times \left(\frac{R_i}{N}\right) \times \left(\frac{C_j}{N}\right) = \frac{R_i \times C_j}{N} \tag{243}$$

where R_i is the i-th row total, C_j is the j-th column total, and N is the grand total. The test statistic is:

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \tag{244}$$

with degrees of freedom:

$$\nu = (r-1)(c-1) \tag{245}$$

7.2 2×2 Contingency Table and Yates' Continuity Correction

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2×2 Contingency Table Formula

For a 2×2 contingency table with counts:

	B_1	B_2	**Total**
A_1	a	b	$a+b$
A_2	c	d	$c+d$
Total	$a+c$	$b+d$	$N=a+b+c+d$

The standard chi-square statistic ($\nu = (2-1)(2-1) = 1$) simplifies algebraically to:

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)} \tag{246}$$

Yates' Correction for Continuity: Because χ^2 is continuous while counts a, b, c, d are discrete integers, Frank Yates introduced a correction when $\nu = 1$ and any cell frequency is small (< 5):

$$\chi^2_{\text{corr}} = \frac{N\left(|ad-bc| - \frac{N}{2}\right)^2}{(a+b)(c+d)(a+c)(b+d)} \tag{247}$$

This continuity correction reduces χ^2_{cal}, preventing inflated Type I error rates.

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Quick Revision — Module 6

Small Samples: t-Test, F-Test, and Chi-Square Distribution | BS-M301

Key Small Sample and Chi-Square Formulas at a Glance

Single Mean t-Test	$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}, \quad s^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2, \quad \nu = n-1$
Two-Sample Mean t-Test	$t = \frac{\bar{x}_1-\bar{x}_2}{s_p\sqrt{1/n_1+1/n_2}}, \quad s_p^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}, \quad \nu = n_1+n_2-2$
Paired t-Test	$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad d_i = x_i - y_i, \quad s_d^2 = \frac{1}{n-1}\sum(d_i-\bar{d})^2, \quad \nu = n-1$
Correlation t-Test	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \quad \nu = n-2$
Fisher's F-Statistic	$F = \frac{s_1^2}{s_2^2} \geq 1, \quad \nu_1 = n_1-1, \quad \nu_2 = n_2-1$
χ^2 Goodness of Fit	$\chi^2 = \sum_{i=1}^{k}\frac{(O_i-E_i)^2}{E_i}, \quad \nu = k-1-m$
2×2 Contingency χ^2	$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}, \quad \nu = 1$
Yates' Correction	$\chi^2_{\text{corr}} = \frac{N(\lvert ad-bc\rvert - N/2)^2}{(a+b)(c+d)(a+c)(b+d)}$

One-Line Definitions

- **Small Sample:** A statistical sample with size $n < 30$, requiring exact distributions (t, F, χ^2) rather than the standard normal Z-test.
- **Degrees of Freedom (ν):** The number of unconstrained, freely variable observations used in evaluating a sample statistic ($\nu = n-k$).
- **Student's t-Distribution:** The exact distribution of the ratio of a standard normal variable to the square root of an independent χ^2 variable divided by its d.f.
- **Pooled Variance (s_p^2):** The weighted average of two independent sample variances used in two-sample t-tests under equal population variances.
- **Paired t-Test:** A test applied to dependent matched pairs where differences $d_i = x_i - y_i$ eliminate extraneous subject variability.
- **Snedecor's F-Distribution:** The distribution of the ratio of two independent chi-square variables, each divided by its d.f. (s_1^2/s_2^2).
- **Chi-Square (χ^2) Distribution:** The distribution formed by the sum of squares of ν independent standard normal random variables.
- **Goodness of Fit:** A statistical test measuring whether an empirical sample frequency distribution adheres to a theoretical distribution.
- **Contingency Table:** A grid matrix displaying the bivariate joint frequency distribution of two qualitative attributes.
- **Yates' Correction:** A continuity correction subtracting $N/2$ from $|ad-bc|$ in 2×2 tables to account for discrete counts in continuous χ^2.

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Frequently Asked / Viva Questions (Part 1)

1. **Q1. Why can we not use the standard normal Z-test for small samples ($n < 30$)?**

   For $n < 30$, the Central Limit Theorem cannot guarantee normality of the sampling distribution unless the parent population is normal. Furthermore, substituting the sample variance s^2 for the unknown σ^2 introduces substantial additional sampling variability, making the tails of the distribution much heavier than $N(0,1)$. Gosset's t-distribution accounts precisely for this extra variability.

2. **Q2. What are the key assumptions underlying the two-sample independent t-test?**

   The assumptions are: (1) both parent populations are normally distributed, (2) the two samples are selected randomly and independently, and (3) the two population variances are equal ($\sigma_1^2 = \sigma_2^2 = \sigma^2$, known as homoscedasticity). Equality of variances is checked using an F-test.

3. **Q3. When should a paired t-test be used instead of an independent two-sample t-test?**

   A paired t-test must be used when the two sets of observations are intrinsically dependent, such as measurements on the same physical subjects before and after a treatment, or twin studies. It isolates the treatment effect by converting paired data into single differences $d_i = x_i - y_i$, thereby eliminating between-subject biological or physical variance.

4. **Q4. State the relation between Student's t, Snedecor's F, and Chi-square χ^2 distributions.**

   Key interrelations include:

   - The square of a t-variable with ν d.f. is an F-variable with $(1, \nu)$ d.f.: $t_\nu^2 = F_{1,\nu}$.
   - As the denominator degrees of freedom $\nu_2 \to \infty$, $\nu_1 F_{\nu_1,\nu_2} \to \chi^2(\nu_1)$.
   - As degrees of freedom $\nu \to \infty$, Student's $t(\nu) \to N(0,1)$.

5. **Q5. Why is the larger sample variance always placed in the numerator in an F-test?**

   By convention, setting $F = s_1^2/s_2^2$ with $s_1^2 \geq s_2^2$ ensures $F \geq 1$. This allows the use of standard upper-tail F-distribution critical value tables without needing to invert lower-tail critical values ($F_{1-\alpha} = 1/F_\alpha$).

6. **Q6. What are the fundamental validity conditions for Karl Pearson's Chi-square test?**

   The conditions are: (1) total sample size $N = \sum O_i \geq 50$, (2) each theoretical expected frequency $E_i \geq 5$ (if $E_i < 5$, adjacent cells must be pooled), (3) individual observations are drawn randomly and independently, and (4) $\sum O_i = \sum E_i = N$.

7. **Q7. How are degrees of freedom calculated in an $r \times c$ contingency table?**

   The degrees of freedom are $\nu = (r-1)(c-1)$. This arises because the marginal row totals impose r constraints and the column totals impose c constraints, with one common grand total constraint N, leaving $(r-1)(c-1)$ independently assignable cells.

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Frequently Asked / Viva Questions (Part 2)

8. **Q8. What is Yates' correction for continuity and when is it mandatory?**

   Yates' correction adjusts the discrete cell counts in a 2×2 table to match the continuous theoretical χ^2 density by subtracting $N/2$ from the absolute difference $|ad - bc|$. It is mandatory when $\nu = 1$ and any cell frequency is less than 5 ($E_{ij} < 5$), preventing spurious rejections of H_0.

9. **Q9. How do you test whether an observed sample correlation coefficient $r = 0.6$ from $n = 18$ pairs is statistically significant?**

   We test $H_0 : \rho = 0$ using $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.6\sqrt{16}}{\sqrt{1-0.36}} = \frac{0.6\times 4}{0.8} = 3.00$. With $\nu = 18 - 2 = 16$ d.f., the critical value at $\alpha = 0.05$ is $t_{0.025,16} = 2.12$. Since $|t_{\text{cal}}| = 3.00 > 2.12$, the correlation is statistically significant.

10. **Q10. What does a non-significant result in a χ^2 goodness-of-fit test signify?**

    When $\chi^2_{\text{cal}} \leq \chi^2_{\text{crit}}$, we fail to reject H_0. This indicates that the observed sampling discrepancies from theoretical frequencies are small enough to be attributed purely to chance, confirming the theoretical model fits the empirical data well.

Summary Matrix: Comparison of Small Sample Tests (t, F, χ^2)

Test	Objective	Test Statistic & Parameters	d.f. (ν)
Single Mean t	**Test** $H_0 : \mu = \mu_0$ (σ **unknown**)	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$, $s^2 = \dfrac{\sum(x_i - \bar{x})^2}{n-1}$	$n - 1$
Two-Sample t	**Compare two means** ($\sigma_1 = \sigma_2$)	$t = \dfrac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}}$, pooled s_p^2	$n_1 + n_2 - 2$
Paired t	**Before vs. After on same units**	$t = \dfrac{\bar{d}}{s_d/\sqrt{n}}$, differences $d_i = x_i - y_i$	$n - 1$
Correlation t	**Test** $H_0 : \rho = 0$	$t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	$n - 2$
Fisher's F	**Compare** σ_1^2 **vs** σ_2^2	$F = \dfrac{s_1^2}{s_2^2}$ ($s_1^2 \geq s_2^2$)	$(n_1 - 1, n_2 - 1)$
Pearson's χ^2	**Goodness of Fit**	$\chi^2 = \sum \dfrac{(O_i - E_i)^2}{E_i}$	$k - 1 - m$
Contingency χ^2	**Independence of attributes**	$\chi^2 = \sum\sum \dfrac{(O_{ij} - E_{ij})^2}{E_{ij}}$ or 2×2 Yates	$(r-1)(c-1)$

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